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MIL-STD-1832

15 March 91

MILITARY STANDARD
DISKETTE MESSAGE FILE FORMATS
FOR
DEFENSE MESSAGING

AMSC N/A AREA IPSC

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FOREWORD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD).

2. This standard, together with the other military documents referenced in Section 2, provides the means for establishing the standardization of messages written to diskette storage media.

3. If there is any conflict between the text and information contained in the Tables or Appendices, the text applies.

4. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technology Integration Center (TIC/TIST) Scott Air Force Base, Illinois 62225-6001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Purpose. The purpose of this standard is to achieve diskette interoperability across Department of Defense (DOD) messaging systems by establishing standard diskette message file formats.

1.2 Application. The requirements of this standard apply to all Departments and Agencies of the DOD using, developing, supporting, or otherwise involved with DOD messaging systems which employ messages residing on magnetic diskette storage media.

1.3 Restrictions. This standard is restricted for use with the MS-DOS 2.0 or higher Microcomputer Operating System (or equal) and its associated diskette format scheme and file naming conventions.

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-1037 - Glossary of Telecommunications Terms

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

a. Joint Army, Navy, Air Force Publication (JANAP) 128 - Automatic Digital Network (AUTODIN) Operating Procedures.

b. Defense Communication Agency (DCA) Circular 370-D175-1 - Defense Communication Service (DCS) AUTODIN Interface and Control Criteria (DSSCS).

c. DSSCS Operating Instructions (DOI)-103 - DSSCS Operating Instructions System/Data Procedures (c).

2.2 Non-Government publications. None.

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence. The examples presented throughout this standard are intended for guidance only. Therefore, the correctness of the information contained in them must be validated through the cited reference(s) prior to actual use in Defense messaging. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

a. ACP - Allied Communication Procedure/Publication

b. AIG - Address Indicator Groups

c. ASCII - American Standard Code of Information Interchange

d. AUTODIN - Automatic Digital Network

e. CIC - Content Indicator Code

f. CRC - Cyclic Redundancy Check

g. CR-LF - Carriage Return-Linefeed

h. DD LMF - Variable length language media format

i. DAG - DSSCS Address Groups

j. DCA - Defense Communication Agency

k. DCS - Defense Communication Service

l. DD - Double Density

m. DDI - Delivery Distribution Indicator

n. DIA - Defense Intelligence Agency

o. DOD - Department of Defense

p. DODISS - Department of Defense Index of Specifications and Standards

q. DOI - DSSCS Operating Instruction

r. DSSCS - Defense Special Secure Communication Systems

s. DTG - Date Time Group

t. EFTO - Encrypted for Transmission Only

u. EM - End of message

v. FED-STD - Federal Standard

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w. FL - Format Line

x. GENSER - General Service

y. GMT - General Message Title

z. HD - High Density

aa. IBM - International Business Machines

ab. IPSC - Information Processing Standards for Computers

ac. JANAP - Joint Army, Navy, Air Force Publication

ad. LMF - Language Media Format

ae. MS-DOS - Microsoft-Disk Operating System

af. NATO - North Atlantic Treaty Organization

ag. OSRI - Originating Station Routing Indicator

ah. PLA - Plain Language Address

ai. RI - Routing Indicator

aj. SARAH - Standard Automated Remote to AUTODIN Host

ak. SD - Single Density

al. SIC - Subject Indicator Code

am. SPECAT - Special Category

an. SSIC - Standard Subject Indicator Code

ao. SSN - Station Serial Number

ap. TCC - Transmission Control Code

aq. TOC - Table of Contents

ar. TRC - Transmission Release Code

3.2 Diskette Volume Label. A descriptive or identifying word or phrase about the information contained on the message disk--the requirements of which are prescribed by MS-DOS Version 2.0 or higher (otherwise known as a volume identification label).

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3.3 File Format. One that is prescribed by MS-DOS Version 2.0 or higher and limited to the American Standard Code of Information Interchange (ASCII) character set.

3.4 Filename. A naming convention prescribed by MS-DOS Version 2.0 or higher.

3.5 MS-DOS. A proprietary (Microsoft) disk operating system (the interface between applications software and the hardware) designed for use with IBM Compatible Personal Computers.

3.6 Message Disk. Magnetic storage media, in the form of a 5.25 or 3.5 inch single density (SD), double density (DD), or high density (HD) floppy diskette, formatted under MS-DOS Version 2.0 or higher.

3.7 Prologue. A set of attribute fields preceding the text of a message file defining the communications characteristics for processing purposes.

3.8 Table of Contents. A file generated by applications software (not MS-DOS) which contains information about the entire message diskette and each individual message contained on it.

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4. GENERAL REQUIREMENTS

4.1. The Message Disk.

4.1.1. Message Disk Definition. Message Disks must first be formatted under MS-DOS version 2.0 or higher and must include a diskette volume label electronically recorded during the disk format process. Individual services and agencies will determine the actual contents of the label. The diskette can contain two types of files: One Table Of Contents (TOC) file and zero or more message files. The TOC file describes the Message Disk and lists the important attributes of the messages it contains. Each message on the disk is contained in its own message file.

4.1.2. Message Disk Creation. The security level for the disk is established at creation time. There should be no existing files or sub-directories on the disk. A TOC file named "Diskette.TOC" will be written in the root directory. If files or sub-directories are added to the Message Disk outside of the implementing system, the implementing system will signal the Message Disk as invalid.

4.1.3. File Formats. All files written to the disk are restricted to the ASCII character set. All language media format (LMF), except variable length record messages (DD LMF) (DCA Circular 370-D175-1), will consist of carriage return-linefeed (CR-LF) sequence. Two forms of an end-of-file condition can exist. In either case, fill characters are used to blank the compliment of the cache buffer disk segment. Non-DD LMF message files are terminated by one SUB (hex 1A) character followed by SUB fill characters to the end of the segment. For DD LMF message files, text records (FL12I) are terminated by the end-of-message (EM) character (hex 19) (CR, LF, and SUB characters are treated as any other text character). The Prologue of the message file will be in the same format as non-DD LMF messages. The end-of-file is signified by an EM character followed by (even-parity) DEL fill characters (hex FF). These characters are used to fill the buffer up to the remaining unwritten segment boundary. (MS-DOS floppy segment size is 512 bytes.) The end of valid information in the file is determined by searching in reverse from the end of the last record in the file and locating the first occurrence of an EM. The fill characters are ignored and used only to ensure no residual information is attached to the message in the last sector during the MS-DOS file write to floppy disk. Specific information concerning Prologue Message Formats and Non-Prologue Message Formats is contained in paragraphs 5.2.1 and 5.2.2 respectively. Specific information regarding use of Defense Special Secure Communication Systems (DSSCS) only fields can be found in DOI-103.

4.1.4. CRC Checks. Unauthorized changes and data errors are detected by the use of Cyclic Redundancy Checks (CRC). Each message on a Message Diskette has a CRC calculated on it which is stored in the TOC file. Before message transmission, this value shall be checked to ensure the message has not changed. The TOC file itself also has a CRC which is recalculated and checked each time the TOC is processed. If this check fails, the Message Disk will be

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signaled as an invalid Message Disk. The CRC algorithm itself is discussed in Appendix A. Appendix Q contains information about the Defense Intelligence Agency (DIA) message level CRC used in DSSCS.

4.1.5. Message Validation. Software validation of messages will be in accordance with the individual requirements of the service/agency. Validation is ensuring that the format rules of this document, and applicable security requirements, are followed.

4.1.6. Security. To reduce the probability of a security compromise, no message with a security level exceeding the security level of the Message Disk (indicated in the TOC file Header) will be allowed to be saved on the Message Disk. In addition, all applicable security directives will be followed. Security considerations apply to both transmit and receive disks.

4.2. The Releasing Document. The releasing document will accompany each diskette delivered to the communication center for transmission. The format and minimum content are contained in Appendix R.

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5. DETAILED REQUIREMENTS

5.1 Table Of Contents Format. The TOC file consists of two major parts: the "TOC Header" and the "TOC Entries." The TOC Entries section is the list of messages after the TOC header. The TOC header contains information about the entire diskette and the TOC entries contain information about each individual message, 1 line per message. The list of messages must always be in order of descending precedence (highest precedence at the top, oldest first within precedence levels).

5.1.1. TOC Header. An example of the TOC Header format is in Appendix B.

5.1.1.1. TOC Header Conventions. All Field Names are left-justified and must appear uppercase and spelled as shown (with underscores). All Field Names are followed by one data line (except the last one: "MESSAGES_FOLLOW"). Data lines contain a colon (:) in column 1 and have a maximum length of 80 characters including the colon. The first two fields (FORMAT_NAME and TOC_CRC), and the last field (MESSAGES_FOLLOW) must appear in the order shown. The in between fields may appear in any order. Software systems will find fields based upon the title of the field, not the order or location. Optional fields may contain valid data, be left empty, or the field and its data line may be left out entirely. The data line of an optional field can be made empty in one of two ways: (1) by putting a colon (:) in the first position of the line, followed by the appropriate number of spaces for that data line, followed by the CR-LF sequence; (2) by following the colon immediately with the CR-LF. Every included field must be followed by one data line (even if it is empty).

5.1.1.2. TOC Header Field Definitions. The following are detailed descriptions of data line entries for each field in the TOC Header:

a. "FORMAT_NAME". This must be the first field in the TOC. The field must begin with the string "TABLE OF CONTENTS", followed by the revision of the standard under which the TOC was created. For example, if the standard is revision A, then the field would appear as: ":TABLE OF CONTENTS/1832A 03 December 1997". MANDATORY Field.

b. "TOC_CRC". This must be the second field in the TOC. Contains 5 numeric ASCII characters representing the CRC calculated on the TOC file itself. The field CRC calculation shall be done on the entire TOC file, both the TOC Header and the TOC Entries. The calculation should be done in accordance with the Appendix A (CRC Algorithm). This field shall be space filled to contain 5 spaces during the actual calculation of the CRC. If the result of the CRC calculation is less than 5 characters, the value shall be right justified in the 5 character field, preceded by spaces. Example: ": 3492". MANDATORY Field.

c. "ORIGINATING_SYSTEM". The name and version indicator of the system that created the Message Diskette. Example: ":SARAH V7.01". MANDATORY Field.

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d. "CLASSIFICATION_OF_DISKETTE". Two character field. Two redundant characters representing the maximum classification message the diskette may contain. Example: ":SS". MANDATORY Field.

e. "COMMUNITY_OF_DISKETTE". Two character field. Two characters representing the communities of traffic the diskette may contain. For DSSCS/GENSER traffic, Y=DSSCS R=GENSER. Example: ":RR", ":RY", or ":YY". MANDATORY Field.

f. "CREATION_DATE". Creation date of the Message Diskette. If included, required format is: YYMMDD. Example: ":890310" for 10 Mar 89. OPTIONAL Field.

g. "CREATION_TIME". Creation time of the Message Diskette. If included, required format is: hhmmssHH. Example ":07124683" for 0712, 46 seconds, and 83 hundredths. OPTIONAL Field.

h. "DISKETTE_RELEASING_OFFICIAL". Identity of releasing official for Message Diskette. Example: ":GEORGE BOSS, GM13". MANDATORY Field.

i. "ORGANIZATION". Identity of originating organization for Message Diskette. Include organization name, office symbol, and telephone number. Example: ":TOP NOTCH SQUADRON, TNS/ME, 123-4567", MANDATORY Field.

j. "COMMENTS". Any comments the originating organization desires to make to the communication center personnel (or other comments). Example would be special message instructions. OPTIONAL Field. A possible comment might be:

":FILENAME--- TYPE- PR CL SP ACTN CRC-- DTG--------- SSN- TOF/TOR CIC- INTERNAL--".

k. "MESSAGES_FOLLOW". This must be the last line of the TOC Header and an exception to the rule that field titles must be followed by a colon on the next line. The next line shall be the first TOC Entry (The TOC Entry for the highest precedence message on the Message Diskette), followed by all other messages in precedence order. MANDATORY Field.

5.1.2. TOC Entries. The following is an example of a complete TOC. The TOC Entries begin after the MESSAGES_FOLLOW line. In this example, all of the optional fields are left out of the TOC header. The ruler under the TOC is not part of the TOC, it is for column reference of the TOC Entries. Conventions and field definitions for the TOC entries follow the example.

```
FORMAT_NAME
:TABLE_OF_CONTENTS/V2
TOC_CRC
:40193
ORIGINATING_SYSTEM
:SARAHv7.01
CLASSIFICATION_OF_DISKETTE
:UU
```


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```
COMMUNITY_OF_DISKETTE
:RR
DISKETTE_RELEASING_OFFICIAL
:GEORGE BOSS, GM13
ORGANIZATION
:TOP NOTCH SQUADRON, TNS/ME, 123-4567
MESSAGES_FOLLOW
MESSAGE.1     U1392 RR UU    MORE 13946                              ZYUW
MESSAGE.2     P1392 RR UU    REDY 78209                              ZYUW
EQIP-ACS.CF1  DD173 RR UU    TXED 19658 211212ZJAN88 0212141
TSR           DD173 RR UU    TXED  7977 221327ZJAN89 0222158        DJBT
MSG0013A      DT128 RR UU    REDY 15110                              ZYUW

--3456789012345678901234567890123456789012345678901234567890123456789
--       1         2         3         4         5         6
(ruler)
```

5.1.2.1. TOC Entry Conventions. There will be one TOC Entry for each message on the Message Disk, one TOC Entry per line. The first one will be on the line immediately after the "MESSAGES_FOLLOW" line. TOC Entries will be ordered by precedence with the highest precedence first. If more than one message of a precedence level is present, order these so the oldest of the precedence is first. The last line of the TOC file is the TOC Entry representing the lowest precedence message on the Message Disk. Each TOC Entry line is broken up into specific fields. Each TOC Entry follows the same format. Starting from the left, the first seven fields, through the CRC field, are mandatory fields. To the right of these are optional fields. The mandatory fields must contain valid data. Optional fields may be spaced, but if filled, they also must contain valid data as specified below. Line terminators (CR-LF) may directly follow the last field used on the line. Information in the TOC Entries fields must match the corresponding message prologue fields (if a corresponding field exists and field data is present) (e.g., The contents of the Content Indicator Code (CIC) field, in columns 66-69, must match the CIC field of the message prologue of the corresponding message, if the field was included in the prologue.)

5.1.2.2. TOC Entry Field Definitions. The following information is provided for explanatory purposes only. Use the specified reference to validate the entries contained in these examples. DISKETTE.TOC file's individual TOC entries look like this:

Columns	Contents
1 - 12	Filename, MS-DOS compatible. (MANDATORY FIELD) The MS-DOS filename and extension (if present) of a message must be contained on this diskette. The filename and extension must be separated by a period. The filename DISKETTE.TOC is reserved.

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Columns	Contents
13	Space
14 - 18	Message Type (5 Alphanumeric) (MANDATORY FIELD) The Prologue Message Types in Table 5.I are allowed. Non-Prologue Message Types in Table 5.II are allowed.
19	Space
20 - 21	Message Precedence (2 Alphas) (MANDATORY FIELD) This is the precedence of the message. It should be utilized to ensure that messages are listed in order of descending precedence. Message Precedences Allowed (See JANAP 128 to validate): "WW" -- CRITIC (DSSCS Only) "YY" -- Emergency Command Precedence "ZZ" -- Flash "OO" -- Immediate "PP" -- Priority "RR" -- Routine
22	Space
23 - 24	Message Classification (2 Alphas) (MANDATORY FIELD) Message Classifications Allowed (See JANAP 128 to validate): "TT" -- Top Secret "SS" -- Secret "CC" -- Confidential "RR" -- Restricted "EE" -- Unclass EFTO "UU" -- Unclass
25	Space

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Columns	Contents

26 - 27 Special Category (SPECAT) Indicator (2 Alphas) (MANDATORY FIELD)

SPECAT Codes Allowed (See JANAP 128 to validate):

" "	-- spaces if non-SPECAT message
"AA"	-- TOP SECRET SIOP-ESI
"BB"	-- ALL OTHER SPECAT MATERIAL
"FF"	-- see Note 1
"LL"	-- see Note 2
"PP"	-- see Note 3
"YY"	-- see Note 4

NOTE:

1. Classified US originated message traffic addressed to activities of the United Kingdom that contain the special handling designators "US-UK EYES ONLY", "DISCREET", or "CHURCH PIKE" must contain 'FF' in the SPECAT field.

2. Messages containing the special handling designator, "ATOMAL", addressed to certain North Atlantic Treaty Organization (NATO) activities and member nations must contain 'LL' in the SPECAT field.

3. Messages containing the special handling designator, "EXCLUSIVE", addressed to certain NATO activities and member nations must contain 'PP' in the SPECAT field.

4. Messages containing the special handling designator, "CRYPTOSECURITY", addressed to certain NATO activities and member nations must contain 'YY' in the SPECAT field.

28 Space

29 - 32 Message Action Field (4 Alphas) (MANDATORY FIELD)

Action Entries allowed are (See JANAP to indicate):

"REDY"	-- Ready for Transmission. Only messages marked REDY will be transmitted.
"TXED"	-- Transmitted message. Message was "acked" by receiving end.
"RECD"	-- Received Message
"BACK"	-- Backup of Message
"REJT"	-- Message Rejected by AUTODIN Host
"ERRO"	-- Error Occurred
"CANX"	-- Canceled

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Columns	Contents
	"BADX" -- Bad Message, failed validation or conversion to JANAP 128
	"MORE" -- Message needs more input to be made ready to Transmit. An example of a MORE message would be a message that is ready except it needs an RI or a Date Time Group (DTG). The operator must add the missing information before attempting to transmit the message. After he adds the information, this field will change to "REDY".
33	Space
34 - 38	Message Cyclic Redundancy Check (CRC) (MANDATORY Field)
	The CRC-16 cyclic redundancy check is calculated and updated at the time the message file is saved to the Message Disk, right-justified and space-filled to the left. Use the same CRC algorithm used to calculate the TOC_CRC in the TOC Header. This is a message integrity check to ensure that the file has not been modified since it was put on the disk. The CRC-16 algorithm is discussed in Appendix A.
39	Space
40 - 51	DTG (12 Alphanumeric) (OPTIONAL Field).
	This is the DTG taken from the Date Time Group Field of the message, in the following format:
	ddhhmmZMMMYY where dd = day, hh = hour, mm = minute, Z = Z (Zulu), MMM = month, YY = year
	for example, 1208Z on 8 Mar 89 = "081208ZMAR89"
52	Space
53 - 56	Station Serial Number (SSN) (4 Numerics) (OPTIONAL Field)
	The SSN number assigned to the message.
57	Space
58 - 64	Time of Transmission (TOT)/Time of Receipt (TOR) (7 Numerics) (OPTIONAL Field)

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Columns	Contents

This field is space filled except for messages with an action of TXED, RECD, REJT, and CANX. For TXED, CANX, and REJT, this is the TOT/attempted transmission of the message, which would normally be taken from the File Time field of the message. For RECD it would represent the time the message was received by the terminal.

The format is as follows :

jjjhhmm where jjj = julian date, hh = Zulu hour, mm = minute
for example, "1080518"
jjjhhmm

Update right after a message has completed (or attempted) transmission or reception.

65 Space

66 - 69 CIC (4 Alphabetic, or 3 Alphabetic and 1 Alphanumeric) (OPTIONAL Field)

CIC from the message.

70 - 80 Native System Use (11 characters) (OPTIONAL Field)

This field is for unique service/agency use.

TABLE 5-I. Prologue message types/LMF reference

Message Type	LMF	Explanation
U1392	Cx/SC	Unprocessed (DD 1392) Data Pattern, no RIs
P1392	Cx/SC	Processed (DD 1392) Data Pattern, with RIs
NR128	xy	Narrative message format
DT128	CC/SC	Data (card) message format
SN128	xy	Narrative service message format
SD128	CC	Data (card) service message
PL128	Cx	Piloted data message format
DD173	.yx	DD173 message in prologue format
VL128	DD	Variable Record Length (see DCA Circular 370-D175-1)
NR103	xy	DSSCS narrative message format (DSSCS Only)
N103S	xy	DSSCS special message format (DSSCS Only)

NOTE: x represents either a C, T, A, or space
y represents either a T, A, or space

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TABLE 5-II. Non-prologue message types/LMF reference

Message Type	LMF	Explanation
RX128	xx/DD	Received (from AUTODIN) JANAP 128 message
NA128	xx/DD	Fully formatted JANAP 128 message (from a user, destined for AUTODIN. This includes modified ACP126.)
RX103	xx	Received DOI-103 message (DSSCS Only)
R103S	xx	Received DOI-103 special message (DSSCS Only)

NOTE: x represents either a C, T, or A

5.2. Message Formats.

5.2.1. Prologue Message Formats. The prologue format was developed to make identification of information contained in it easy. Each field of information has a field name line followed by 1 or more lines of data representing that field. Each data line starts with a colon (:) in column 1, as a delimiter.

```
Example:  line 1  --  FORMAT_NAME
          line 2  --  :NR128
          line 3  --  LMF
          line 4  --  :CA
          line 5  --  C_MESSAGE
          line 6  --  :N
          line 7  --  INTERNAL_MESSAGE_CRC
          line 8  --  : 2245
          line 9  --  RELEASERS_NAME
          line 10 --  :
          line 11 --  OFFICE_SYMBOL
          line 12 --  :SKA
          --
          --
          --
```

See Appendices for complete examples of prologued messages.

5.2.1.1. Message Prologue Conventions. All Prologue Field Names are left-justified and must appear uppercase and spelled as shown (with underscores). All Prologue Field names are followed by one or more data lines. Data lines contain a colon (:) in column 1 and have a maximum length of 80 characters, including the colon. The Prologue Field Name "TEXT_LINES_FOLLOW" terminates the prologue and is followed by message text. The "FORMAT_NAME" field name must be the first field name and the "TEXT_LINES_FOLLOW" field must be the last field name of a prologue. The fields in between may appear in any order. Software systems will find fields based upon the title of the field, not the order or location. Mandatory fields must be present and contain valid data in the data line in accordance with the description or the reference. In some

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cases, valid data may be an empty entry (space-filled field or colon followed by CR-LF). Optional fields may contain valid data, be left empty, or the field and its data line may be left out entirely. The data line of an optional field can be made empty in one of two ways: (1) by putting a colon (:) in the first position of the line, followed by the appropriate number of spaces for that data line, followed by the CR-LF sequence; (2) by following the colon immediately with the CR-LF. Every included field must be followed by one data line (even if it is empty). Alpha characters in data lines are always uppercase, except in the "COORDINATION_MEMO" field. Lines that do not have a colon (:) in position 1 are field names. Multi-line fields of data can contain from 1 up to the maximum number of lines for that field. The FL12I is appended to the end of the prologue following the last field's data line (See paragraph 4.1.3). The end of text for the message is determined by the end-of-file mark (SUB character). (No EM is attached to the file.) (See paragraph 4.1.3). Character sets are limited by LMF requirements defined in DCA Circular 370-D175-1.

5.2.1.2. Message Prologue Field Requirements. Table 5-III shows the required prologue field information for each type of message that can be passed on the communication diskette. For clarity, and a better understanding of field content, some slots have entries for additional information. In Table 5-III, the matrix column headers identify appropriate format, LMF and message type combinations. In Table 5-I, the LMF field type column shows the LMF combinations involved for each FORMAT_NAME type (C is card, A is narrative, D is variable length magnetic tape, T for teletype, or space for terminal default). The C_MESSAGE field row identifies the message type to be either S for service, N for normal, or P for pilot for all message format types where applicable. The fields listed in this table are in the recommended field order. Table 5-III is a guidance tool for practitioners. In the event of a conflict between entries seen in this table and the text of this standard, the text of this standard shall take precedence.

TABLE 5-III. Field utilization by message type

The following table describes the field utilization required or allowed for each message type possible in the standard diskette format.

NOTE: x represents C, A, T, or space. y represents A, T, or space.
Also, U1392 same as P1392, except RI is not required.

LEGEND			
M	- Mandatory	(N)	- normal
Opt	- Optional	(S)	- Service
M/Opt	- Conditional statement, see field description	(P)	- Pilot
[empty block, field is not used]			

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TABLE 5-III. Field utilization by message type - Continued.

Field Name	Message Type																	
																	OSCS ONLY	
	U1392/P1392	U1392/P1392	U1392/P1392	U1392/P1392	U1392/P1392	U1392/P1392	U1392/P1392	DT128	NR128	DT128	SN128	SD128	FL128	FL128	VL128	DD173	NR108	N108S
FORMAT_NAME	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M
UAF	M (Oy)	M (CC)	M (Oy)	M (CC)	M (Oy)	M (CC)	M (SC)	M (SC)	M (xy)	M (CC)	M (xy)	M (CC)	M (Oy)	M (CC)	M (CD)	M (yx)	M (xy)	M (xy)
C_MESSAGE	M(N)	M(N)	M(S)	M(S)	M(P)	M(P)	M(N)	M(N)	M(N)	M(N)	M(S)	M(S)	M(P)	M(P)	M(N)		M(N)	M(N)
INTERNAL_MESSAGE_CRC	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt
COORDINATION_MEMO	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt
DISTRIBUTION																Opt		
DRAFTERS_NAME	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt	Opt		
DRAFTERS_OFFICE																Opt		
RELEASERS_NAME	Opt	Opt			Opt	Opt	Opt									Opt		
OFFICE_SYMBOL	Opt	Opt			Opt	Opt	Opt											
SPECIAL_INSTRUCTIONS																Opt		
PAGE_OF																Opt		
DAY_TIME_ZULU	M				M				M				M			M	M	M
MONTH	M				M				M				M			M	M	M
YEAR	M				M				M				M			M	M	M
PRECEDENCE_FIRST	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	Opt	M	M
PRECEDENCE_SECOND (Note 1)	Opt				Opt				Opt				Opt			Opt	Opt	Opt
CLASS	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M

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TABLE 5-III. Field utilization by message type - Continued.

Field Name	Message Type																DSCS ONLY	
	U1392/P1392 (Cy) N	U1392/P1392 (CC) N	U1392/P1392 (Cy) S	U1392/P1392 (CC) S	U1392/P1392 (Cy) P	U1392/P1392 (CC) P	U1392/P1392 (SC) N	DT128 (SC) N	NR128 (xy) N	DT128 (CC) N	SN128 (xy) S	SD128 (CC) S	PL128 (Cy) P	PL128 (CC) P	VL128 (CD) N	DD173 (yx)	NR108 (xy)	NL085 (xy)
SPECAT	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M
CIC	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	Opt	M	M
OAI (Note 2)																Opt		
BOOK																Opt		
MSG_HANDLING																Opt		
FROM																M		
OSRI_PLA	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M		M	M
TO (Note 3)	Opt	M	M	M	Opt	M	M	M	Opt	M	M	M	Opt	M	M	Opt	Opt	Opt
INFO (Note 4)	M/Opt				M/Opt				M/Opt				M/Opt			M/Opt	M/Opt	M/Opt
PLA_RI																Opt		
XMT	Opt				Opt				Opt				Opt			Opt		
ACCT	Opt				Opt				Opt				Opt			Opt		
TRC	Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt	Opt	Opt				
PLA_TEXT	Opt	Opt			Opt	Opt			Opt	Opt			Opt	Opt			Opt	Opt
P_CIC					M	M							M	M				
P_ADDRESSEE_RI					M	M							M	M				
ORIG_ID	Opt	Opt			Opt	Opt	Opt											
REMARKS	Opt	Opt			Opt	Opt	Opt											

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TABLE 5-III. Field utilization by message type - Continued.

Field Name	Message Type																		
	U1392/P1392 (Oy) N	U1392/P1392 (CC) N	U1392/P1392 (Oy) S	U1392/P1392 (CC) S	U1392/P1392 (Oy) P	U1392/P1392 (CC) P	U1392/P1392 (SC) N	DT128 (SC) N	NR128 (xy) N	DT128 (CC) N	SN128 (xy) S	SD128 (CC) S	PL128 (Oy) P	PL128 (CC) P	WL128 ((DD) N	DD173 (yx)	DSSCS ONLY NR108 (xy)	DSSCS ONLY N108S (xy)
FL12A	M		M	M	M				M		M	M	M			M	M	M
FL12B	Opt		Opt	Opt	Opt				Opt		Opt	Opt	Opt			Opt		
FL12B_COMENT	Opt		Opt	Opt	Opt				Opt		Opt	Opt	Opt			Opt		
FL12C	Opt				Opt				Opt				Opt			Opt	Opt	Opt
FL12D	Opt				Opt				Opt				Opt			Opt	Opt	Opt
FL12E_H	Opt				Opt				Opt				Opt			Opt	Opt	Opt
TIME_OF_FILE	Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt	Opt	Opt			Opt	
ORIGINAL_STATION_SERIAL_NUMBER	Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt
DECLASSIFICATION_INSTRUCTIONS	Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt	Opt	Opt		Opt	M/Opt	M/Opt
TRANSMISSION_CONTROL_CODE (DSSCS ONLY)																	M	M
CODEWORDS (DSSCS ONLY)																	Opt	Opt
CAVEATS (DSSCS ONLY)	Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt	Opt	Opt		Opt	Opt	Opt
DIA_MSG_LEVEL_ORC (DSSCS ONLY)	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt	M/Opt
LESS (DSSCS ONLY)																Opt		
RADIR_HEADER	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt	Opt	Opt			Opt	Opt	Opt	Opt
TEXT_LINES_FOLLOW	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M

Notes:

1 The "PRECEDENCE_SECOND" field is required if different from "PRECEDENCE FIRST" field.

2 OMI is mandatory on DD173 format messages processed by systems accredited by the DIA. (DSSCS Only)

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3 The "TO" field consists of two types of information: a RI and the PLA.

The U1392 and DD173 message types are the only ones that do not require the RI to be present with the PLA. (example: ":RUWTABC/TINKER AFB OK//SKAA//")

4 The "INFO" field is required if no "ACTION" field is present. The "INFO" field consists of two types of information, the first is the RI, the second is the PLA. (Example: ":RUWTABC/TINKER AFB OK//SKAA//")

The U1392 and DD173 message types are the only ones that do not require the RI to be present with the PLA.

5.2.1.3. Message Prologue Field Definitions. The following are detailed descriptions of the prologue fields for both basic formats. Allowed field lengths do not include the colon.

a. "FORMAT_NAME". Mandatory for all message types. This 5 character field contains the Message Type, which is associated with specific LMFs, and the state of readiness of the message for processing (e.g., "NR128"). Valid entries are prescribed in Table 5-III.

EXAMPLE: ":NR128"

b. "LMF". Mandatory for all message types. This is a 2 character alpha field (e.g., :CA as a Card to ASCII Narrative format). Note: A space filled field is assumed to be a narrative LMF. See Table 5-I for a listing of legal LMFs.

c. "C_MESSAGE". Mandatory for all message types, except DD173, for which it is not used. This is a 1 character field. It is used to determine the type of JANAP 128 message to generate and the type of data to acquire from the operator. The possible entries are:

"N" for normal JANAP 128 message based on LMF
"S" for a service message in JANAP 128 for narrative messages
"P" for a piloted message in JANAP 128 for card messages

EXAMPLE: ":N"

d. "INTERNAL_MESSAGE_CRC". Optional for all message types. This a 5 character field (columns 2 through 6, after the colon). If this field is present in the prologue, it must contain the message CRC. The CRC is right-justified in the field and space filled to the left, if needed. Reference Table 5-III for more discussion on the CRC.

EXAMPLE: ": 2478"

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e. "COORDINATION_MEMO". Optional for all message types. This is a multi-line field with up to 80 characters per line, no limit on the number of lines. It is for internal use only and will not be transmitted with the message. It can contain coordination information added by reviewers. It is available for pre-transmission coordination only. Recommend the first line state "COORDINATION REQUIRED" or "COORDINATION COMPLETE". This can be a procedural activity that will provide a check for proper and final coordination before processing the message for transmission.

```
EXAMPLE: ":COORDINATION REQUIRED
         :MFR
         :FROM:  CO (Mr Bob Dowright)
         :   These are the comments provided to correct or change
         :   this message. You can enter as many lines of input as
         :   needed to convey the information.
         :
         :MFR
         :FROM:  XP (Mr Checker)
         :   More than one reviewer can use this area to provide
         :   review and correction to the drafter of the message.
         :   Coordination is not limited by the room available here.
         :"
```

f. "DISTRIBUTION". Optional for DD173. Not used by other message types. This is a multi-line field of up to 6 lines. Each line can have up to 69 characters. It can contain drafter or system generated distribution information for the message.

EXAMPLE: ":EID/CC-2"

g. "DRAFTERS_NAME". Optional for all message types. This field is up to 34 characters in length. It is for internal use only and will not be transmitted with the message. It contains the drafters or operators name and title.

EXAMPLE: ":SSGT I. M. Fine"

h. "DRAFTERS_OFFICE". Optional for DD173. Not used by other message types. This field is up to 34 characters. It is for internal use only and will not be transmitted with the message. It contains the office symbol and phone number of the drafter.

EXAMPLE: ":XPT/45410"

i. "RELEASERS_NAME". Optional for some DD 1392 and DD173 message types. This field is up to 32 characters is length. It is for internal use only and will not be transmitted with the message.

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(1) For DD173 message types it contains the releasers name, title, office symbol, and phone number.

EXAMPLE: ":BOSS/OIC/CSD/42925"

(2) For all message types except DD173 it can contain the name and title of the person who is authorized to release messages for transmission.

EXAMPLE: ":JAMES Q. JONES, SUPVR"

j. "OFFICE_SYMBOL". Optional for all DD 1392 message types, except service messages. Not used for DD173 or JANAP 128 message formats. This field is up to 29 alphanumeric characters long. It is for internal use only and will not be transmitted with the message. It is the office symbol and duty phone of the Releasing Officer.

EXAMPLE: ":DOOCN, EXT 45221"

k. "SPECIAL_INSTRUCTIONS". Optional for DD173. Not used by other message types. This is a multi-line field of up to 5 lines. Each line can contain up to 34 characters. It contains special instructions appropriate for the DD173.

EXAMPLE: ":CONFIRM DELIVERY TO ACTION ADDRESS
:TCC, SEND AS A SINGLE ADDRESS
:MESSAGE"

l. "PAGE_OF". Optional for DD173. Not used by other message types. This is an 8 character field. It can contain the page number information for the DD173 form.

EXAMPLE: ":01 01" (would read 01 of 01 on form)
12345678
(The 12345678 is for character position reference only).

m. "DAY_TIME_ZULU". Mandatory. Plaindress, codress, and DD173 messages require this field. Abbreviated plaindress and data pattern do not use this field. The DTG field consists of 7 alphanumeric characters. It is divided into 3 parts. The first part is the current date expressed as 2 digits. If the current date has only one digit, it must be preceded by a 0 (zero). The second part of the field is the ZULU time expressed in 4 digits. The third part consists of 1 character, always a "Z" for ZULU time. This is the time the message was released by proper authenticating authority. This field matches the DTG field in this message's TOC Entry.

NOTE: The times of 0000 and 2400 are invalid times.

If an entry is made in this field, the month and year fields must also be entered for proper DTG entry.

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EXAMPLE: ":010817Z"
ddhhmmZ

n. "MONTH". Mandatory. Plaindress, codress, and DD173 messages require this field, abbreviated plaindress and data pattern do not use this field. Note: If an entry is made in this field, the Day_Time_Zulu and YEAR fields must also be entered for proper DTG entry. This is a 3 character alpha field (uppercase). It is the first three characters of the current month.

NOTE: EXAMPLE: ":MAR"

o. "YEAR". Mandatory. Plaindress, codress, and DD173 messages require this field. Abbreviated plaindress and data pattern do not use this field. Note: If an entry is made in this field, the Day_Time_Zulu and month field must also be entered for proper DTG entry. This is a 2 character numeric field. It consists of the last two digits of the current year.

EXAMPLE: ":89"

p. "PRECEDENCE_FIRST". Mandatory. This is a 2 character alpha field. It is the precedence of the message repeated. This field matches the Message Precedence field in this message's TOC entry. The valid precedences in order of highest to lowest are (See JANAP 128 to validate):

"WW" - CRITIC (DSSCS Only)
"YY" - ECP (Emergency Command Precedence)
"ZZ" - FLASH
"OO" - IMMEDIATE
"PP" - PRIORITY
"RR" - ROUTINE

EXAMPLE: ":PP"

q. "PRECEDENCE_SECOND". Optional. This is a 2 character alpha field. It is used to assign a lower precedence to the info addressees of the message repeated. Valid precedences are the same as in the PRECEDENCE_FIRST.

r. "CLASS". Mandatory for all message types. This is a 2 character alpha field. It is simply the first character of a valid classification repeated. This field matches the Message Classification field in this message's TOC Entry. The valid classifications in the order of highest to lowest are (See JANAP 128 to validate):

"TT" - TOP SECRET
"SS" - SECRET
"CC" - CONFIDENTIAL

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"RR" - RESTRICTED
"EE" - UNCLASSIFIED EFTO
"UU" - UNCLASSIFIED

EXAMPLE: ":UU"

s. "SPECAT". Mandatory for all message types. This is a 5 character alpha field. It is simply the SPECAT Indicator. A space filled entry indicates no SPECAT designation. This field matches the SPECAT Indicator field in this message's TOC Entry. The valid SPECAT Indicators are (See JANAP 128 to validate):

(5 SPACES) - no SPECAT designation
"AAAAA" - TOP SECRET SIOP-ESI
"BBBBB" - ALL OTHER SPECAT MATERIAL
"FFFFF" - see note 1
"LLLLL" - see note 2
"PPPPP" - see note 3
"YYYYY" - see note 4

NOTE: 1. Classified US originated message traffic addressed to activities of the United Kingdom that contain the special handling designators "US-UK EYES ONLY", "DISCREET", or "CHURCH PIKE" must contain 5 'F's in the SPECAT block.

2. Messages containing the special handling designator, "ATOMAL", addressed to certain NATO activities and member nations must contain 5 'L's in the SPECAT block.

3. Messages containing the special handling designator, "EXCLUSIVE" addressed to certain NATO activities and member nations must contain 5 'P's in the SPECAT block.

4. Messages containing the special handling designator, "CRYPTOSECURITY" addressed to certain NATO activities and member nations must contain 5 'Y's in the SPECAT block.

EXAMPLE: ":BBBBB"
EXAMPLE: ": " (NO SPECAT)

t. "CIC". Mandatory for all message types, except for DD173 for which it is optional. This field is 4 alphanumeric, or 3 alpha and 1 alphanumeric characters long. It will tell the communication center personnel if any special requirements for message delivery are required. The CIC must conform to the JANAP 128 format. NOTE: Leave this field space filled for narrative to narrative messages unless a specific CIC has been assigned.

EXAMPLE: ":ZYVW"

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u. "OMI". Optional for DD173. Not used by any other message types. NOTE: This field is mandatory for all DD173 message types processed on systems accredited by DIA (DSSCS Only). This field is 12 alphanumeric characters in length. It is for internal use only and will not be transmitted with the message. This is the Originator/Message Identification. It is the originator's means of identifying his message.

EXAMPLE: ":SKAS 010842"
EXAMPLE: ":XPP 031425 X"

v. "BOOK". Optional for DD173. Not used by any other message types. This is a 3 character alpha field. It will be either "YES" or "NO". If YES, each addressee will receive a copy of the message without knowing who else the message was addressed to. If NO, each addressee will know who has received this message.

Note: An empty entry is assumed to be "NO".

EXAMPLE: ":YES"
EXAMPLE: ":NO "
EXAMPLE: ": "

w. "MSG_HANDLING". Optional for DD173. Not used by any other message types. This field is up to 64 characters long. It is used for any operating signals that may apply to the message. For a DSSCS DD173, this field may contain DDI entries in addition to any operating signals that may apply to the message (DSSCS Only).

EXAMPLE: ":ZYQ"
(for messages with 'deliver as a single address message' in SPECIAL HANDLING INSTRUCTIONS)

x. "FROM". Mandatory for DD173. Not used by any other message types. This is a multi-line field of up to 5 lines. It can contain up to 55 alphanumeric characters per line, as specified by service directives. It is the PLA of the originator of the message. Each PLA can have a maximum of 4 continuation lines. The PLA on any line can extend beyond the 55 character limit onto the next line as a continuation line. PLA continuation lines are identified as follows : A colon in column 1 followed by 4 spaces and then the continuation information. Where office symbols are allowed, the start of office symbol is a double slash. If more than one office originated the message, office symbols are separated by a single slash. The end of office symbols is denoted by a double slash.

```
EXAMPLE:  ":CCSC TINKER AFB OK//SKAA//
           :    CONTINUATION LINE STARTS HERE"
            1234567890123456789012345678901234567890123456789012345
                      1         2         3         4         5
```


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or

EXAMPLE: ":HQ MAC SCOTT AFB IL//DO/LG//"
1234567890123456789012345678901234567890123456789012345
1 2 3 4 5

y. "OSRI_PLA". Mandatory for all message types except DD173, for which it is not used. This is a multi-line field of up to 5 lines. It can contain up to 63 alphanumeric characters per line. Each line has 2 fields of information separated by a slash (/). The first field is a 7 character field used to hold the RI for the originator. The second field is up to 55 characters in length and contains the PLA for the RI. Both fields are left-justified. DSSCS messages will contain only the PLA which will be preceded by 8 spaces. There is no RI entered for a continuation line.

NOTE: For communications terminals, the OSRI must precede the mandatory PLA for each FROM Address.

EXAMPLE: ":RUWTAAA/CCSO TINKER AFB OK
: / CONTINUATION LINE STARTS HERE"
12345678901234567890123456789012345678901234567890123456789012345678901234567890123
1 2 3 4 5 6

z. "TO". Optional/Mandatory. Mandatory for plaindress formatted messages when there is no entry in the "INFO" field. Optional for all other messages.

(1) For DD173 message prologues types:

(a) This field can be up to 55 alphanumeric characters per line, as specified by service directives.

(b) It is the list of PLAs to which the message is to be sent.

(c) The PLA on any line can extend beyond the 55 character limit onto the next line as a continuation line.

(d) Each PLA can have a maximum of 4 continuation lines.

EXAMPLE: ":CCSC TINKER AFB OK"
1234567890123456789012345678901234567890123456789012345
1 2 3 4 5

(e) Where office symbols are allowed, the start of office symbols is a double slash. If more than one office receives the message, office symbols are separated by a single slash. The end of office symbols is denoted by a double slash.

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```
EXAMPLE:  ":HQ MAC SCOTT AFB IL//DO/LG//"
          12345678901234567890123456789012345678901234567890123456789012345
                    1         2         3         4         5
```

(2) For all message types except DD173:

NOTE: Communication terminals have eight characters added to the beginning of each PLA for a RI and slash (/).

(a) This field can be up to 63 alphanumeric characters per line, as specified by service directives.

(b) Each line has 2 fields of information separated by a slash (/). The first field is a 7 character field used to hold the RI for the addressee. The second field is up to 55 characters in length, as specified by service directive, and contains the PLA for the RI. Both fields are left-justified. DSSCS messages will contain only a PLA which will be preceded by 8 spaces.

(c) There is no RI entered for a continuation line.

```
EXAMPLE:  ":RUWTAAA/CCSO TINKER AFB OK
          :       /    CONTINUATION LINE STARTS HERE"
          12345678901234567890123456789012345678901234567890123456789012345678901234
                    1         2         3         4         5         6
```

(d) AIG and General message title (GMT) identifiers can be substituted for a PLA. These special conditions change the requirement for a RI according to the following rules:

1. The AIG/GMT line will not contain an RI.

2. The associated RIs will be entered immediately on the lines following the line with the AIG/GMT. There can be any number of RIs associated with one AIG/GMT.

```
EXAMPLE:  ":COLLECTIVE NUMBER 21              --(FOR DD173 FORMAT)
          :       /COLLECTIVE NUMBER 22       --(FOR J128 AND DD 1392)
          :RUWTABC/
          :RUWTABB/
          :RUWTACC/"
          12345678901234567890123456789012345678901234567890123456789012345
                    1         2         3         4         5
```

NOTE: If the routing indicator's last three letters are "QCT" and a single address, then the message will be transmitted as a Quality Control Test. It will have no filetime and SSN (Filetime = 000000Z, SSN = 0000).

For DSSCS DD173 message prologue type, DSSCS Address Groups (DAG) identifiers can be substituted for a PLA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

aa. "INFO". Optional/Mandatory. Mandatory for plaindress formatted messages when there is no entry in the "TO" field. Optional for all other messages. Format for this field is the same as the "TO" field description. It is for the INFO addressees.

ab. "PLA_RI". Optional for DD173. Not used by any other message types. This is a 7 character alpha field. This field allows Army, Navy, and Marine Corps users to generate messages which require PLA to RI conversion. It contains the "SUU" RI of the telecommunications system which will perform the PLA to RI conversion.

EXAMPLE: ":RUWTSUU"

ac. "XMT". Optional for narrative format message types. Not used by others. This is a multi-line field, with each line having up to 55 alphanumeric characters. It is used to indicate which, if any, PLA addressees contained in collective routes or AIGs are exempt from delivery of this message.

EXAMPLE: ":HQ MAC SCOTT AFB IL"

ad. "ACCT". Optional for narrative format message types. This field can be up to 55 alphanumeric characters in length. It is the billing symbol for authorized users of US military communications facilities for messages which may require commercial means to get to the final destination.

EXAMPLE: ":ARX GMRC"

ae. "TRC". Optional for all message types. Not valid for single card, DD LMF, and DD173. This is a 2 character alpha field. The Transmission Release Code (TRC) consists of two letters that help ID foreign addressees. If only addressed to a single foreign nation, it will be redundant. If addressed to two foreign nations, then the TRC will be in alphabetic order. If more than two addressees, then two or more messages will be required. Valid TRC characters are found in JANAP 128.

EXAMPLE: ":BB"

af. "FL4_TEXT". Optional for normal and pilot message types, not used by DD173. This is a multi-line field, with each line up to 69 alphanumeric characters in length. It may contain special text or instructions that need action by the communications operator, and are normally considered messages to the communications operator. The proper format is found in JANAP 128. This information will be transmitted as FL4 of the message.

```
EXAMPLE:  ":ZFD RUWTABC"
          ":ZOV RUCLGBA2144 REROUTE OF RUEKJCS1234 0011530"
            12345678901234567890123456789012345678901234567890123456789 -->
                     1         2         3         4         5         6 -->
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

ag. "P_CIC". Mandatory for piloted messages. Not used for DD173. This is a 4 character field. This special field is used to capture the CIC to be put into the pilot header of a message. The CIC must conform to the JANAP 128 format.

EXAMPLE: ":ZDKW"

ah. "P_ADDRESSEE_RI". Mandatory for piloted messages. Not used for DD173. This is a special multi-line field of up to 500 lines, maximum of 63 characters per line. Each line is a 7 character field representing an RI followed by a slash (/). The second field is up to 55 characters in length and contains the PLA for the RI. Both are left-justified. This field is for entering the addressees that are to receive the piloted message (See JANAP 128 for specific information).

```
EXAMPLE:  ":RUWTAAA/CCSC TINKER AFB OK"
          ":      /    CONTINUATION LINE STARTS HERE"
            1234567890123456789012345678901234567890123456789012345678901234567890123
                     1         2         3         4         5         6
```

ai. "ORIG_ID". Optional for some message types. Not used for JANAP 128 or DD173. See Table 5-I. This field is 26 alphanumeric characters long. It is the RCS, follow-up, status, etc. This field is for internal use only and will not be transmitted with the message.

EXAMPLE: ":DD 1392 field information."

aj. "REMARKS". Optional for DD 1392 normal, pilot, and single card message types. Not used for DD173 or JANAP 128 messages. This field is 48 alphanumeric characters long. It is for internal use only and will not be transmitted with the message. It can contain any special instructions or remarks about the message the originator may want to send to the operator at the communication center.

EXAMPLE: ":This field used only for DD 1392 information."

ak. "FL12A". Mandatory for narrative, service, and DD173 message types. This field can be variable up to the length of the formal spelling for the CLASSIFIED field as defined in JANAP 128. It contains the message classification, in accordance with JANAP 128. If not required, it is space-filled or empty.

EXAMPLE: ":UNCLAS"
EXAMPLE: ":T O P S E C R E T"

al. "FL12B". Optional for narrative, service, and DD173 message types. This is a variable length field. It contains the message special handling designations as defined in JANAP 128.

EXAMPLE: ":DISCREET"

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Check the source to verify that this is the current version before use.

am. "FL12B_COMMENT". Optional for narrative, service, and DD173 message types. This field is limited in length to the remaining space on a 69 character line, after the classification and special handling designation wording is added. This field is for any information that may pertain to the classification. It contains any special handling instructions not provided by the contents of field "FL12B". Communication terminals may start service information in this field (See JANAP 128 to validate).

EXAMPLE: ":SVC ZUI SVC 0245 RUEDCSA1234 1921600."

an. "FL12C". Optional for normal and pilot narrative and DD173 message types. This is a multi-line field with up to 69 characters on each line. It contains releasability statements or appropriate regional defense organization security classification statement (See JANAP 128 to validate).

EXAMPLE: ":INFORMATION ALLOWED BY PARA 355 ACP 121 US SUPP-1"

ao. "FL12D". Optional for normal and pilot narrative and DD173 message types. This is a multi-line field with up to 69 characters on each line. It contains subject indicator code (SIC), standard subject indicator code (SSIC), and/or DDI (See JANAP 128 to validate).

EXAMPLE: ":SSIC CODE AND INFORMATION."

ap. "FL12E_H". Optional for normal and pilot narrative and DD173 message types. This is a multi-line field with up to 69 characters on each line. It is for general use to allow entry of other FL12 component fields of information as necessary (See JANAP 128 to validate).

EXAMPLE: ":SPECIAL DELIVERY INSTRUCTIONS"
EXAMPLE: ":EXERCISE NAME"
EXAMPLE: ":SEE PARA. 323.b ACP 121-1"
EXAMPLE: ":A. REFERENCE ONE"

aq. "TIME_OF_FILE". Optional for all message types, except DD173, single card, and variable length messages, for which it is not used. This is a field of 7 numeric characters. The format of the data is Julian date, hour, minute (jjjhhmm).

EXAMPLE: ":0761315"

ar. "ORIGINAL_STATION_SERIAL_NUMBER". Optional for all message types, except DD173, DD LMF, and single card messages, for which it is not used. This is a field of 4 numeric characters. Its use is to capture the original sender's generated SSN for transmission purposes.

EXAMPLE: ":0233"

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Check the source to verify that this is the current version before use.

as. "DECLASSIFICATION_INSTRUCTIONS". Mandatory for classified messages, excluding DD LMF and single card message types. This field is omitted for unclassified messages. This information is required by applicable security documents. This is a multi-line field. Field length is determined by LMF type. There is no limit to the number of lines (See JANAP 128 to validate).

EXAMPLE: ":DECLASSIFY AFTER 120 DAYS"

at. "TEXT_LINES_FOLLOW". Mandatory for all message types. This field terminates the prologue and signals the beginning of text (FL12I). Text lines are not preceded by a colon. The length of each line is dependent on the LMF of the message being processed.

NOTE: If LMF indicates single card, the text will be 1 line with a maximum of 41 characters.

EXAMPLE: "THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT"
"IT IS NOT REQUIRED TO BE PRECEDED BY A COLON."

au. "TRANSMISSION_CONTROL_CODE". (DSSCS Only) Mandatory Field for all DSSCS message types, except DD173. This is a 3 character alpha field. The Transmission Control Code (TCC) consists of 3 letters which allow the AUTODIN network to control delivery of the message. The valid TCCs are found in DOI-103.

av. "CODEWORDS". (DSSCS Only) Optional field for all DSSCS message types. This is a multi-line field with up to 69 characters on each line. It contains Special Security Designations for appropriate handling of DSSCS traffic as defined in DOI-103 or other applicable security directives.

EXAMPLE: ":XXXXXX YYYYYYY ZZZZZZ"

aw. "CAVEATS". (DSSCS Only) Optional for all DSSCS message types. This is a multi-line field with up to 69 characters per line. It contains Special Handling Instructions as defined in DOI-103 or other applicable security directives.

EXAMPLE: ":HANDLE VIA XXXX CHANNELS"

ax. "DIA_MSG_LEVEL_CRC". (DSSCS Only) Optional for all message types. NOTE: This field is mandatory for all messages processed on systems accredited by DIA. This is a 4 character alphanumeric field. If the field is present in the prologue, it must contain the DIA Message Level CRC. The CRC value is a 4 character ASCII representation of a 4 digit hexadecimal value. Only the numerals 0-9 and the letters A-F are valid. Reference Appendix Q for more discussion of the DIA Message Level CRC.

EXAMPLE: ":2C7F"

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Check the source to verify that this is the current version before use.

ay. "LESS". (DSSCS Only) Optional for all DSSCS narrative format message types. Not used by other message types. This is a multi-line field, with each line having up to 55 alphanumeric characters. It is used to indicate which, if any, PLAs contained in DAGs are exempt from delivery of this message.

EXAMPLE: ":SSO YYYYY"

az. "RADDR_HEADER". Optional for narrative, service, and DD173 message types. Only for a readdressal message, this represents all sets of format lines 5 through 10 of the message being readdressed. This is a multi-line field of up to 69 characters per line.

5.2.2. Non-Prologue Message Formats. The message types "NA128" (native 128), "RX128" (receive 128), "RX103" (receive DOI-103), and "R103S" (receive DOI-103 Special) are the only non-prologue message types allowed. They are JANAP 128, DOI-103, and DOI-103 special messages consisting of FLs 2 through 16. All lines will be terminated with a CR-LF to accommodate the MS-DOS file structures. No blank lines will be present. Modified ACP 126 formatted messages fall under the NA128 message type. Modified ACP 126 is essentially a subset of JANAP 128.

EXAMPLE of a NA128/RX128 file format:

```
RTTUZYUW RUEBABA1234 3201400-UUUU--RUKKLAA.
ZNR UUUUU
R 181239Z NOV 89
FM USS JUNEAU
TO CNO WASHINGTON DC
BT
UNCLAS
TEXT
BT
#1234
NNNN
```

NOTE: The SUB character will be inserted on the line immediately below FL15 or FL16, or on the line immediately following the distribution lines, if present. It shall be on a line by itself.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

EXAMPLE of a RX103 file format. (DSSCS Only)

```
RAAMZYUW YYYYYY 0001 1231234-MNSH--YXXXXX.
ZNY MMNSH
ZKZK PP XXX DE
P 031225Z MAY 89
FM AAAAA
TO VVVVV
INFO CCCCCC
ZEM
UNCLAS EFTO
QQQQ
THIS IS AN EXAMPLE MESSAGE.
#0001
NNNN
```

EXAMPLE of a R103S file format. (DSSCS Only)

```
ZCZCCSN001
PP YXXXXX
DE YZZZZZ #1234 2501440
ZNY MMNSH
ZKZK PP XXX DE
P 191722Z DEC 89
FM SITE1
TO SITE2
ZEM
C L A S S I F I C A T I O N  CODEWORDS AND SPECIAL HANDLING
INSTRUCTIONS MAY TAKE SEVERAL LINES
QQQQ
TEST OF MESSAGE........
#1234
NNNN
```


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Check the source to verify that this is the current version before use.

6. NOTES

6.1 MS-DOS is a registered trademark of Microsoft Corporation.

6.2 IBM is a registered trademark of International Business Machines Incorporated.

6.3 Norton Utilities are copyrighted by Peter Norton.

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APPENDIX A

CRC-16 ALGORITHM

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION.

40.1 Method of Calculation. The CRC is calculated on a "whole file" basis. Fill characters are never included. For all files, CR-LF line delimiters and SUB file delimiters are excluded from the calculation, with the following exception: For the text portion of DD LMF messages (FL12I), EMs are the delimiters and are thus excluded from the calculation. CR, LF, and SUB characters are included in the calculation, since they are treated as normal text characters in this case. If the file being calculated contains the field where the CRC value will be placed, then this field should be space filled during the calculation. This is true for the TOC_CRC field of the TOC file and the INTERNAL_MESSAGE_CRC field of message files.

40.2 Examples. This appendix contains examples of the CRC calculation done in four different languages and a line-by-line calculation of the CRC of sample messages and TOC. The following are provided for your information:

a. The languages are:

(1) Ada-like pseudo code.

(2) Ada (three versions).

(3) Pascal (two versions).

(4) C (two versions).

b. Notes about CRC efficiency:

(1) Most of these examples demonstrate the algorithm, but not in its most efficient form. The algorithm can be speeded up considerably by table referencing and even more through partial use of assembly language.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

(2) As an example, the Ada version, as shown, took 61 seconds to calculate the CRC on a large test file (55 KB). By utilizing a 16 byte table, the time required was reduced to 9.6 seconds (6.3 times faster); using 256 byte table, it took 5.6 seconds (10.8 times faster); by coding part of the routine in assembly language and using the larger table, it took 0.4 seconds (152.5 times faster). The second Ada example uses a table and assembly language.

c. The first Pascal example and the first Ada example were actually compiled and run with the described test file to ensure that they came up with the same results. The test file, for the CRC calculation used, contained the following information:

ABCDEFGHIJKLMNOPQRSTUV
WXYZ. 1234567890 abcde
fghijklmnopqustuvwxyz.

d. The following are listings from viewing the test file with Norton Utilities:

Norton Utilities -- Hex format

```
 TEST.FIL sectors 26-27 in hex format       Cursor at offset 0, hex 0
41424344 45464748 494A4B4C 4D4E4F50 51525354 55560D0A ABCDEFGHIJKLMNOPQRSTUV
5758595A 2E203132 33343536 37383930 20616263 64650D0A WXYZ. 1234567890 abcde
66676869 6A6B6C6D 6E6F7071 72737475 76777879 7A2E1A03 fghijklmnopqrstuvwxyz.
```

Hex representation | ASCII characters

Norton Utilities -- Text format

TEST.FIL sectors 26-27 shown in text format

ABCDEFGHIJKLMNOPQRSTUV<CR><LF>
WXYZ. 1234567890 abcde<CR><LF>
fghijklmnopqrstuvwxyz.<end-file-marker>

NOTE: The <CR><LF> notation was added. The special graphics characters that show on the screen for CR-LF were not printable in this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

40.2.1 Ada Examples.

40.2.1.1 Ada Like Pseudo-code Example.

```
procedure CRC_Buffer (IN Buffer; IN Length; IN OUT CRC_Accumulator)is
     begin
          Poly := 120001(8);
          For i := 1 to Length loop
               Letter := Buffer(i);
               For Bit := 1 to 8 loop
                    Shift1 := Letter rem 2;
                    Shift2 := CRC_Accumulator rem 2;
                    CRC_Accumulator := CRC_Accumulator / 2;
                    if Shift1 /= Shift2
                      then
                        CRC_Accumulator := CRC_Accumulator XOR Poly;
                    end if;
                    Letter := Letter / 2;
               end for;
          end for;
     end CRC_Buffer;
```

40.2.1.2 First Ada Example.

```
with Sys_Bufr, Unsigned, Unchecked_Conversion, text_io;
use  Sys_Bufr, Unsigned, text_io;
procedure Return_Crc(file      : in  Buf_Recs_Ptr_Type;
                     CRC_Value : out Long_integer) is

      function Convert_to_Byte is new Unchecked_Conversion(
                                    Source => character,
                                    Target => byte);

    Poly   : constant word := 8#120001#;
    Line   : Buf_Recs_Ptr_Type;
    Letter : byte;
    Shift1 : byte;
    Shift2 : byte;
    CRC_Accumulator : word := 0;
    End_Of_List : boolean;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
    begin

    Line := Get_First_Buffer_In(File);

    for i in 1..Size_Of(File) loop -- This loop for line-by_line

      for i in 1..Line.Count - 1 loop -- This loop for
                                      -- character-by-character

        Letter := Convert_to_Byte(Line.Text(i));
        for bit in 1..8 loop
          Shift1 := Letter rem 2;
          Shift2 := byte(CRC_Accumulator rem 2);
          CRC_Accumulator := CRC_Accumulator / 2;

          if Shift1 /= Shift2 then
            CRC_Accumulator  := CRC_Accumulator xor Poly;
          end if;
          Letter := Letter /2;

-- debug code
     Put("bit:" & integer'image(bit) & "    ");
     put("    Shift1:" & byte'image(shift1));
     put("    Shift2:" & byte'image(shift2));
     Put_line("    CRC_Accumulator:" & word'image(CRC_Accumulator));
-- debug code
        end loop;

      end loop;
-- debug code
     Put("line:" & integer'image(i) & "    ");

     Put_line("The interim CRC Value is :" &
                                        word'image(CRC_Accumulator));

-- debug code

      Get_Next_Buffer(Line,End_Of_List); -- Get another line

    end loop;

    CRC_Value := Long_integer(CRC_Accumulator);

end Return_Crc;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
-------------------------------------------------

with return_crc, sys_bufr, text_io, read_file;
use   sys_bufr, text_io;

procedure prt_crc is

   Input     : string(1..12) := (others => ' ');
   File      : Buf_Recs_Ptr_Type;
   CRC_Value : long_integer;
   length    : integer;

    begin

       Put ("Enter Filename : ");
       Get_Line(Input,length);
       Read_file(Input,File);
       Return_CRC(File,Crc_Value);

       Put_Line("The CRC Value is :" & long_integer'image(CRC_Value));

    end prt_crc;
```

Results of Running Ada Version with TEST.FIL

```
-----------------------------------------

A:\ADA_VER>prt_crc
Enter Filename : test.fil
line: 1    The interim CRC Value is : 15418
line: 2    The interim CRC Value is : 260
line: 3    The interim CRC Value is : 29553
The CRC Value is : 29553
```

40.2.1.3 Second Ada Example.

Here is the CRC done in Ada using a table and an assembly language interface:

```
with System;
with Unsigned, Sys_Bufr;
use  Sys_Bufr, Unsigned;
package body CRC_Package is

  type CRC_Table_256 is array (0..255) of Word;

  CRC_Table    : CRC_Table_256;   -- Table of remainder values.
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
POLY         : Constant Word := 8#120001#;  -- This is the
                                            -- polynomial for
                                            -- the CRC-16 calculation;
CRC_Value    : Word;  -- contains the CRC

                      -- interface to assembly routine
procedure CRC_String_256 (CRC_Table_Start: IN     System.Address;
                          String_Start   : IN     System.Address;
                          String_Count   : IN     Natural;
                          CRC_Value      : IN OUT Word);
pragma Interface (Ada, CRC_String_256);
procedure Generate_Table_256 is

   Result_Word : Word:= 0;

begin

  for i in 0..255 loop -- Loop thru 256 times, to create
                       --                  each table entry

    Result_Word:= word(i);

    for i in 1..8 loop -- Loop thru 8 times for each
                       -- Result_Word to process each bit!

      if (Result_Word and 1)  =  1 then --Odd number in Result_Word!
        Result_word:= (Result_Word / 2)  XOR Poly;
      else           -- ^ -----------^ Shift right one bit and XOR
                     --                 with the constant Poly_Word!
        Result_Word:= Result_Word / 2; --Even number in Result_Word!
      end if;       -- ^ -----------^ Shift right one bit!  --

    end loop;

      CRC_Table(i):= Result_Word; -- Install answer into the table!

  end loop;
end Generate_Table_256;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
Procedure CRC_String_16 (File        : IN     Buf_Recs_Ptr_Type;
                         CRC         :    OUT Integer) is
                        ------------------------------------------
                        -- Module Ident:  11.8.1
                        --
                        -- This module calculates the Cyclic Redundancy
                        -- Code for a file which is a Buf_Recs_Ptr_Type
                        -- and returns a CRC which is an integer.  For
                        -- a  good discussion of CRCs,  see  "Computer
                        -- Networks",  by Andrew S. Tanenbaum, Prentice
                        -- Hall, Inc. Englewood Cliffs, New Jersey, Jan
                        --  1961
                        -- Here  is  a  quick idea  of  what  is  being
                        -- calculated:  The  CRC  is the residual  from
                        -- division  of  the  data stream  by  the  CRC
                        -- polynomial.  The data stream is also thought
                        -- of  as a polynomial,  with the highest order
                        -- term being the lowest bit of the first  byte
                        -- of the data stream and the lowest order term
                        -- of  the polynomial being the high bit of the
                        -- last byte of data.  The actual division  is
                        -- performed  on  the  data  stream  polynomial
                        -- multiplied  by X^y where y is the degree  of
                        -- the CRC polynomial.  The CRC is then stored
                        -- in  the sixth TOC header buffer.  When  the
                        -- receiver  gets  the file,  the CRC  is  again
                        -- computed  and matched against  the  original
                        -- CRC  which was stored in the sixth SARAH TOC
                        -- header buffer. If the two do not match, then
                        -- the files must have been altered.
                        ------------------------------------------

  Temp_Word          : Word := 0;
  Buffer             : Buf_Recs_Ptr_Type;
  Local_File         : Buf_Recs_Ptr_Type;
  End_Of_List        : Boolean;

begin

  CRC_Value := 0;  -- must initialize CRC value for each new file
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
    Local_File:= File;

                        -- Go thru the file buffer by buffer, and for each
                        -- buffer, call the assembly routine
                        -- which calculates
                        -- the accumulative CRC value for each new buffer
    Buffer:= Get_First_Buffer_In(Local_File);

    for i in 1..Size_Of(Local_File) loop -- This loop for line-by_line

                        -- interface to assembly
        CRC_String_256 ( CRC_Table'ADDRESS,
                         Buffer.Text'ADDRESS,
                         Buffer.Count - 1,
                         CRC_Value);

      Get_Next_Buffer(Buffer,End_Of_List); -- Get another line

    end loop;

    CRC:= to_integer(CRC_Value); -- Finally, the answer!!!

  end CRC_String_16;
begin

   Generate_Table_256;  -- build the CRC table at elaboration time

end CRC_Package;
```

The following is the assembly routine that goes with the above Ada routine.

```
;*************************************************************************
;                                                                        *
;          CRC_STRING_256                                                *
;                                                                        *
; This routine calculates a CRC based on the CRC-16 algorithm. It       *
; relies on a table of 256 words set up prior to being called. It takes*
; the address of this table from the stack, as well as the address of  *
; the string (buffer) of characters to be processed, the count (length)*
; of the string), and the initial CRC value. It puts the new CRC value *
; back on the stack before exiting, unless the count is zero, in which *
; case we just exit because the CRC won't have changed                  *
;                                                                        *
;*************************************************************************
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
 .286C
CRC_SEG     SEGMENT BYTE PUBLIC
            PUBLIC  CRC_STRING_256
            ASSUME  CS:CRC_SEG

CRC_STRING_256 PROC FAR

                                ; Alsys convention
            PUSH    BP
            MOV     BP,SP
                                ; save registers
            PUSH    ES
            PUSH    DS
            PUSHA
                                ; Take data off the stack:
            MOV      CX,[BP+14] ; Count (number of characters in string)
            JCXZ     CRC_DONE   ; if count is zero, exit
            LDS      SI,[BP+6]  ; ds:si contain address of CRC_Table
            LES      DI,[BP+10] ; es:di contain address of string
            MOV      AX,[BP+16] ; initial CRC value

NEXT:                           ; Loop through the string,
                                ;  one byte at a time:
            XOR      AL,ES:[DI] ; ax (CRC) <= CRC xor next byte
            INC      DI         ; point at next byte in string
            XOR      BX,BX      ; bx <= 0
            XCHG     AL,BL      ; bx <= ax (CRC) and 00FFh
            XCHG     AL,AH      ; ax <= ax (CRC) shr 8
            SHL      BX,1       ; bx <= bx + bx
            XOR      AX,[BX+DS:SI] ; CRC <= (CRC shr 8) xor
                                ;          CRC_Table[CRC and 00FF]
            LOOP     NEXT       ; count <= count - 1

            MOV      [BP+16],AX ; save the new CRC value

CRC_DONE:
                                ; restore registers
            POPA
            POP      DS
            POP      ES
            POP      BP
                                ; pop 10 bytes off stack,
                                ;           leaving BP+16 and +17
                                ; which contain the new CRC value.
                                ; (get this value from the .lst file
                                ; of the Ada compile)
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
          RET     10
CRC_STRING_256 ENDP
CRC_SEG   ENDS
          END
```

40.2.1.4 Third Ada Example.

The following listing was developed and compiled using the Alsys Ada compiler.

```
with Sys_Bufr, Unsigned, Unchecked_Conversion, text_io;
use  Sys_Bufr, Unsigned, text_io;
procedure Return_Crc(file      : in  Buf_Recs_Ptr_Type;
                     CRC_Value : out Long_integer) is

    function Convert_to_Byte is new Unchecked_Conversion(
                                    Source => character,
                                    Target => byte);

  Poly   : constant word := 8#120001#;
  Line   : Buf_Recs_Ptr_Type;
  Letter : byte;
  Shift1 : byte;
  Shift2 : byte;
  CRC_Accumulator : word := 0;
  End_Of_List : boolean;

  begin

  Line := Get_First_Buffer_In(File);

  for i in 1..Size_Of(File) loop -- This loop for line-by_line

    for i in 1..Line.Count - 1 loop -- This loop for character-by-character

      Letter := Convert_to_Byte(Line.Text(i));
      for bit in 1..8 loop
        Shift1 := Letter rem 2;
        Shift2 := byte(CRC_Accumulator rem 2);
        CRC_Accumulator := CRC_Accumulator / 2;

        if Shift1 /= Shift2 then
          CRC_Accumulator := CRC_Accumulator xor Poly;
        end if;
        Letter := Letter /2;
      end loop;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
      end loop;
-- debug code
     Put("line:" & integer'image(i) & "    ");

     Put_line("The interim CRC Value is :" & word'image(CRC_Accumulator));
-- debug code

      Get_Next_Buffer(Line,End_Of_List); -- Get another line

    end loop;

    CRC_Value := Long_integer(CRC_Accumulator);

end Return_Crc;
```

40.2.2. PASCAL Example.

40.2.2.1 First PASCAL Example.

```
PROGRAM CRC ;

{ This utility prompts a user for a filename, then opens the file for
  input, and performs a CRC check on the file.

  This utility was compiled with Turbo Pascal v3.01A.
}

CONST
   POLY_Seed : INTEGER = $A001 ;

TYPE
   File_String_T            = STRING [ 81 ] ;
   CRC_Accumulator_Value_T  = INTEGER ;

PROCEDURE CRC_GENERATING_ALGORITHM
                       ( File_String : File_String_T
              VAR CRC_Accumulator_Value : CRC_Accumulator_Value_T ) ;

VAR
   Bit_Indx                : INTEGER ;
   File_Line_Index         : INTEGER ;
   First_Shift_Value       : INTEGER ;
   Second_Shift_Value      : INTEGER ;
   Character_Ordinal_Value : INTEGER ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
BEGIN

   FOR File_Line_Index := 1 TO LENGTH ( File_String )
   DO
    BEGIN
    Character_Ordinal_Value :=
                  INTEGER ( File_String [ File_Line_Index ] ) ;

    FOR Bit_Indx := 1 TO 8
    DO
       BEGIN
       First_Shift_Value     := Character_Ordinal_Value MOD 2 ;
       Second_Shift_Value    := ABS ( CRC_Accumulator_Value MOD 2 );
       CRC_Accumulator_Value := CRC_Accumulator_Value SHR 1 ;

       IF First_Shift_Value <> Second_Shift_Value
       THEN
          BEGIN
          CRC_Accumulator_Value :=
                              CRC_Accumulator_Value XOR POLY_Seed ;
          END ; { IF First_Shift_Value <> Second_Shift_Value }

       Character_Ordinal_Value := Character_Ordinal_Value SHR 1 ;
       END ; { FOR Bit_Indx := 1 TO 8 }
    END ; { FOR File_Line_Index := 1 TO LENGTH ( File_String ) }
END ; { PROCEDURE CRC_GENERATING_ALGORITHM }

VAR
   Bit_Indx              : INTEGER ;
   Mask_Value            : INTEGER ;
   Input_File_Line_Count : INTEGER ;
   Real_Additive         : REAL ;
   CRC_Real_Value        : REAL ;
   Input_File            : TEXT ;
   CRC_String            : STRING [ 5 ] ;
   File_Path_And_Name    : STRING [ 50 ] ;
   Input_Text_Line       : File_String_T ;
   CRC_Accumulator_Value : CRC_Accumulator_Value_T ;

BEGIN

  CRC_Accumulator_Value := 0 ;
  WRITE ( 'ENTER FILENAME : ' ) ;
  READLN ( File_Path_And_Name ) ;
  ASSIGN ( Input_File , File_Path_And_Name ) ;
  RESET ( Input_File ) ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
Input_File_Line_Count := 0 ;

REPEAT
   Input_File_Line_Count := Input_File_Line_Count + 1 ;
   READLN ( Input_File , Input_Text_Line ) ;
   CRC_GENERATING_ALGORITHM
                    ( Input_Text_Line , CRC_Accumulator_Value ) ;

   WRITELN ( 'File Line = ' , Input_File_Line_Count:3 ,
             '   ' , 'CRC Value = ' , CRC_Accumulator_Value ) ;
UNTIL ( EOF ( Input_File ) ) ;

WRITELN ( 'After Final Pass, CRC value = ' , CRC_Accumulator_Value );

Mask_Value    := $0001 ;
Real_Additive := 1.0 ;
CRC_Real_Value:= 0.0 ;

IF CRC_Accumulator_Value < 0
THEN
   { This block of code converts a negative INTEGER number
     ( CRC_Accumulator_Value ), into a positive REAL number
     ( CRC_Real_Value ).

     IF  The CRC_Accumulator_Value is a negative number, then
     the  value  needs to be converted into a  positive  value.
     This  will only be necessary if the  CRC_Accumulator_Value
     is  stored  in a 16 bit signed variable type ( INTEGER  ).
     If  the  CRC_Accumulator_Value can be stored into  16  bit
     unsigned  variable  type,  then  this block  will  not  be
     needed.


   BEGIN

   FOR Bit_Indx := 1 TO 16
   DO
      BEGIN

      IF ( CRC_Accumulator_Value AND Mask_Value ) = Mask_Value
      THEN
         BEGIN
         CRC_Real_Value := CRC_Real_Value + Real_Additive ;
         END ; { IF ( CRC_Accumulator_Value AND Mask_Value ) <> 0 }
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
        Real_Additive := Real_Additive * 2.0 ;
        Mask_Value := Mask_Value SHL 1 ;

        END ; { FOR Bit_Indx := 1 TO 16 }
      END ; { IF CRC_Accumulator_Value > -1 }
  STR ( CRC_Real_Value:5:0 , CRC_String ) ;

  WRITELN ;
  WRITELN ( 'INTEGER = ' , CRC_Accumulator_Value ) ;
  WRITELN ( '   REAL = ' , CRC_Real_Value:5:0 ) ;
  WRITELN ( ' STRING  >' , CRC_String , '<' ) ;
  WRITELN ( '          12345' ) ;

END .
```

Results of Running Pascal Version with TEST.FIL

```
A:\>crc_p
ENTER FILENAME : test.fil
File Line =   1   CRC Value = 15418
File Line =   2   CRC Value = 260
File Line =   3   CRC Value = 29553
After Final Pass, CRC value = 29553

INTEGER = 29553
   REAL =     0
 STRING  >    0<
          12345
```

40.2.2.2 Second PASCAL Example.

```
PROGRAM CRC ;

{ This utility prompts a user for a filename, then opens the file for  input,
  and performs a CRC check on the file.

  This utility was compiled with Turbo Pascal v3.01A.
}

CONST
   POLY_Seed : INTEGER = $A001 ;

TYPE
   File_Line_T            = STRING [ 81 ] ;
   CRC_Accumulator_Value_T  = INTEGER ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
VAR
   Ans                     : STRING [1] ;
   Output_File             : TEXT ;
   Debug_File              : TEXT ;
   Print_Line_Count        : INTEGER ;

PROCEDURE CRC_GENERATING_ALGORYTHM (     File_Line   : File_Line_T ;
            VAR CRC_Accumulator_Value : CRC_Accumulator_Value_T ) ;

VAR
   Bit_Indx                : INTEGER ;
   File_Line_Index         : INTEGER ;
   First_Shift_Value       : INTEGER ;
   Second_Shift_Value      : INTEGER ;
   Character_Ordinal_Value : INTEGER ;

BEGIN

   FOR File_Line_Index := 1 TO LENGTH ( File_Line )
   DO
      BEGIN
      Character_Ordinal_Value := INTEGER
                           ( File_Line [ File_Line_Index ] ) ;

      FOR Bit_Indx := 1 TO 8
      DO
         BEGIN

         First_Shift_Value     := Character_Ordinal_Value MOD 2 ;
         Second_Shift_Value    := ABS ( CRC_Accumulator_Value MOD 2 ) ;

  { The following 3 lines provide debugging information. }
         IF Ans <> 'N'
           THEN
           WRITE ( Debug_File ,  'Acc1 = ' , CRC_Accumulator_Value:6 ) ;

         CRC_Accumulator_Value := CRC_Accumulator_Value SHR 1 ;

  { The following 3 lines provide debugging information. }
         IF Ans <> 'N'
         THEN
           WRITE ( Debug_File ,  '  Acc2 = ' , CRC_Accumulator_Value:6 ) ;

         IF First_Shift_Value <> Second_Shift_Value
         THEN
            BEGIN
            CRC_Accumulator_Value := CRC_Accumulator_Value XOR POLY_Seed ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
  { The following 3 lines provide debugging information. }
            IF Ans <> 'N'
               THEN
               WRITE ( Debug_File ,  '  Acc3 = ' , CRC_Accumulator_Value:6 ) ;

            END  { IF First_Shift_Value <> Second_Shift_Value }

  { The following 4 lines provide debugging information. }
          ELSE
            IF Ans <> 'N'
               THEN
               WRITE ( Debug_File , '                ' ) ;

          Character_Ordinal_Value := Character_Ordinal_Value SHR 1 ;

  { The following 10 lines provide debugging information. }
          IF Ans <> 'N'
          THEN
             BEGIN
             WRITELN ( Debug_File ,
                     '  Ch_Ord  = ' , Character_Ordinal_Value:2
                     , '  Shf 1   = ' , First_Shift_Value
                     , '  Shf 2   = ' , Second_Shift_Value ) ;
             WRITE ( 'Output Trace Values? (Y/N) ' ) ;
             READLN ( Ans ) ;
          END ; { IF Ans <> 'N' }

          END ; { FOR Bit_Indx := 1 TO 8 }
       END ; { FOR File_Line_Index := 1 TO LENGTH ( File_Line ) }
END ; { PROCEDURE CRC_GENERATING_ALGORYTHM }

VAR
   Space_Indx            : INTEGER ;
   Bit_Indx              : INTEGER ;
   Mask_Value            : INTEGER ;
   Input_File_Line_Count : INTEGER ;
   CRC_Real_Value        : REAL ;
   Input_File            : TEXT ;
   CRC_String            : STRING [ 5 ] ;
   File_Path_And_Name    : STRING [ 50 ] ;
   Output_File_Name      : STRING [ 50 ] ;
   Debug_File_Name       : STRING [ 50 ] ;
   Input_Text_Line       : File_Line_T ;
   TOC_CRC_Line          : File_Line_T ;
   Prior_Line            : File_Line_T ;
   CRC_Accumulator_Value : CRC_Accumulator_Value_T ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
BEGIN

   CRC_Accumulator_Value := 0 ;
   Prior_Line := '' ;
   Debug_File_Name  := 'CRC.BUG' ;

   ASSIGN ( Debug_File , Debug_File_Name ) ;
   REWRITE ( Debug_File ) ;
   Output_File_Name  := 'CRC.LST' ;
   ASSIGN ( Output_File , Output_File_Name ) ;
   REWRITE ( Output_File ) ;

   WRITELN ( Output_File ) ;
   WRITELN ( Output_File ) ;
   WRITELN ( Output_File ) ;

   WRITE ( 'ENTER FILENAME : ' ) ;
   WRITE ( Output_File , 'ENTER FILENAME : ' ) ;
   READLN ( File_Path_And_Name ) ;
   WRITELN ( Output_File ,  File_Path_And_Name ) ;

   WRITELN ( Output_File ) ;

   WRITE  ('Output Trace Values? (Y/N) ' ) ;
   READLN (Ans) ;
   ASSIGN ( Input_File , File_Path_And_Name ) ;
   RESET ( Input_File ) ;

   Input_File_Line_Count := 0 ;

   REPEAT
      Input_File_Line_Count := Input_File_Line_Count + 1 ;
      READLN ( Input_File , Input_Text_Line ) ;

      IF ( File_Path_And_Name = 'DISKETTE.TOC' )
               AND ( Prior_Line  =  'TOC_CRC' )
         THEN
            BEGIN
            TOC_CRC_Line  :=  ':     ' ;
            CRC_GENERATING_ALGORYTHM ( TOC_CRC_Line
                                     , CRC_Accumulator_Value ) ;
         END { IF Prior_Line  =  'TOC_CRC' }

         ELSE

            CRC_GENERATING_ALGORYTHM ( Input_Text_Line
                                     , CRC_Accumulator_Value ) ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
    CRC_Real_Value := CRC_Accumulator_Value ;
    IF CRC_Real_Value < 0

       THEN
       CRC_Real_Value := CRC_Real_Value + 65536.0 ;
    STR ( CRC_Real_Value:5:0 , CRC_String ) ;

    WRITE ( Output_File ,  'Line ' , Input_File_Line_Count:3 ,
            '  >', Input_Text_Line, '<' ) ;

    IF ( File_Path_And_Name = 'DISKETTE.TOC' )
              AND ( Prior_Line  =  'TOC_CRC' )

       THEN
       BEGIN
          FOR Space_Indx := 1 TO 26
          DO
             WRITE ( Output_File ,  ' ' ) ;

          WRITE ( Output_File ,
                'Blank CRC during calculation     >:     <' ) ;
          FOR Space_Indx := 1 TO 13
          DO
             WRITE ( Output_File ,  ' ' ) ;
       END

    ELSE
       FOR Space_Indx := 1 TO ( 86 - LENGTH ( Input_Text_Line ) )
       DO
           WRITE ( Output_File ,  ' ' ) ;
    WRITELN ( Output_File ,    'Interim CRC Value = ' , CRC_String ) ;

  Prior_Line  :=  Input_Text_Line ;

UNTIL ( EOF ( Input_File ) ) ;

CRC_Real_Value := CRC_Accumulator_Value ;
IF CRC_Real_Value < 0
   THEN
   CRC_Real_Value := CRC_Real_Value + 65536.0 ;
STR ( CRC_Real_Value:5:0 , CRC_String ) ;

WRITELN ( Output_File ) ;
WRITELN ( Output_File ,  'After Final Pass, CRC value = >'
                       , CRC_String , '<' ) ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
   CLOSE ( Debug_File  ) ;
   CLOSE ( Output_File ) ;

END .
```

40.2.3. C Examples.

40.2.3.1 First C Example. This variation of the CRC algorithm is used to CRC newly created TOCs, to CRC newly generated message files destined for output, to verify the CRC of TOCs received as input, and to verify the CRC of message files received as input.

```
#include <stdio.h>

#define POLY  0120001          /* Initial seed for CRC */
#define TOC_CRC              "TOC_CRC"
#define SPACE_5              ":    "
#define DISKETTE_TOC             "diskette.toc"
#define NBR_BYTES                      83
#define LINE_FEED            '\n'
#define CARR_RET             '\r'
#define DOS_EOF              '\x1A'
#define NULL_CHAR            '\0'

unsigned long CRC_Value;         /* Calculated CRC Value */
Return_Crc (char * f_name)
{

FILE *fopen(), *msg_file;
char          *status;

/*************************************************************************
 The magic number 83 occurs because of Harris idiosyncrasies...this
 buffer size will not harm any file based on a record size of 80.
 *************************************************************************/

char          buffer[NBR_BYTES];
char          *Line = &buffer[0];
int           k = 0;
int           i;
int           bit;
int                     crc_flag = 0;
```


Source: https://assist.dla.mil -- Downloaded: 202610-26T07:54Z
Check the source to verify that this is the current version before use.

```
/*****************************************************************************
 The flag substitute_flag is set so only the TOC will be scanned for the
 key phrase TOC_CRC.  This eliminates the need to scan every line of
 every file.
 *****************************************************************************/

int                         substitute_flag = 0;
unsigned char  Letter;
int            Shift1;
int            Shift2;
unsigned long  CRC_Accumulator = 0;

/* open the file and get the first buffer of data */

    msg_file = fopen (f_name, "r");

/*****************************************************************************
 If the file cannot be opened, return -1 to the calling program.
 *****************************************************************************/

        if (msg_file == NULL) {
                return(-1);
        }

    status = fgets (Line, NBR_BYTES, msg_file);

/*****************************************************************************
 If this is the TOC file, set the substitute_flag.
 *****************************************************************************/

        if ((strstr(f_name, DISKETTE_TOC)) != NULL)
              substitute_flag = 1;

    while (status != NULL) {

/*****************************************************************************
 If substitute_flag is set (= 1) and the buffer contains the phrase
 "TOC_CRC", set a flag (crc_flag).
 *****************************************************************************/

        if (substitute_flag == 1) {

                        if ((strstr(Line, TOC_CRC)) != NULL) {
                                crc_flag = 1;
                }
        }
        i = 0;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
/*****************************************************************************
 Not all records in a file contain the CR-LF sequence or "\n".  Fgets
 places a NULL ("\0") at the end of the buffer when a record is read. This
 NULL is the trigger to stop here CRCing the record.
 *****************************************************************************/

        while ((Letter=Line[i++]) != NULL_CHAR) {

/*****************************************************************************
 fgets converts CR-LF ('\r\n') sequences into a lone LF, but to be sure,
 prevent all such characters, including the MS-DOS EOF (hex 1A), from being
 included in the CRC calculation.
 *****************************************************************************/

                    if ((Letter != LINE_FEED) && (Letter != CARR_RET) && (Letter != DOS_EOF)) {

                for (bit = 1; bit <= 8; bit++) {
                                Shift1 = (int) (Letter & 1);
                                Shift2 = (int) (CRC_Accumulator & 1);
                                CRC_Accumulator = CRC_Accumulator / 2;
                            if (Shift1 != Shift2) {
                                CRC_Accumulator = (CRC_Accumulator ^ POLY);
                    } /* end if */
                    Letter = Letter / 2;
              } /* for */
          } /* end if */
      } /*while*/
      status = fgets (Line, NBR_BYTES, msg_file);

/*****************************************************************************
 A new record has been read and the CRC_flag is set:  replace any data
 after the colon with 5 spaces and CRC the record; reset the CRC_flag,
 reset the substitute_flag.
 *****************************************************************************/

        if (crc_flag == 1) {
                        memcpy(Line, SPACE_5, sizeof(SPACE_5) - 1);
                        crc_flag = 0;
                        substitute_flag = 0;
                }
    } /*while*/
    (void) close (msg_file);
    return (CRC_Accumulator);
}
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

40.2.3.2 Second C Example.

```
#include <stdio.h>

#define POLY    0120001                         /* Initial seed for CRC    */

char              f_name(9)="A:173.AT"          /* FileName                */
unsigned    long        CRC_Value;              /* Calculated CRC Value    */

main ()
  {
  CRC_Value = (long) Return_Crc();
  printf ("%u\n",CRC_Value);
  }

Return_Crc ()
   {
FILE            *msg_file;
char            *status;
char            buffer[83];
char            *Line = &buffer;
int             nbr_bytes = 82;
int             k = 0;
int             i;
int             bit;
unsigned char   Letter;
int             Shift1;
int             Shift2;
unsigned long   CRC_Accumulator = 0;

/* Open the file and get the first buffer of data */

   msg_file = fopen (f_name, "r");
   status = fgets (Line, nbr_bytes, msg_file);
   while (status != NULL)
      {
      i = 0;
      while ((Letter=Line[i++]) != '\n')
         {
         for (bit = 1; bit <= 8; bit++)
            {
            Shift1 = (int) (Letter & 1);
            Shift2 = (int) (CRC_Accumulator & 1);
            CRC_Accumulator = CRC_Accumulator / 2;
            if (Shift1 != Shift2)
              {
              CRC_Accumulator = (CRC_Accumulator ^ POLY) ;
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
        }
      Letter = Letter / 2;

    }
  }
  printf ("line: %i  crc: %u\n", ++k, CRC_Accumulator) ;
  status = fgets (Line, nbr_bytes, msg_file) ;
  }
(void) close (msg_file) ;
return (CRC_Accumulator) ;
}
```

Results Running the C Version with TEST.FIL

```
A:\C_VERSIO>CRC
line: 1  crc: 15418
line: 2  crc: 260
line: 3  crc: 13892
13892
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

40.2.4 TOC/DD173/Debug Examples.

The following listing provides information about the TOC file to aid in debugging the CRC program code. The line number is provided to aid in knowing the line number being checked. The leading ">" points to the first valid character in the line and indicates that the character following the arrow is included in the CRC calculation. The leading ">" also indicates the character that is in character position 1 for that line of the sample file. The trailing "<" marker indicates that a CR-LF is in that column position and points to the last valid character in the line for computation of the CRC.

40.2.4.1 CRC Listing of Sample TOC File.

```
ENTER FILENAME : T_MS1WS4.ASC

Line   1  >FORMAT_NAME<                                        Interim CRC Value =  9506
Line   2  >:TABLE OF CONTENTS<                                 Interim CRC Value = 45359
Line   3  >TOC_CRC<                                            Interim CRC Value = 18180
Line   4   >:44797<   Blank CRC during calculation >:  <       Interim CRC Value =  1059
Line   5  >ORIGINATING_SYSTEM<                                 Interim CRC Value = 46925
Line   6  >:CNTCINST 2300.X<                                   Interim CRC Value = 61904
Line   7  >CLASSIFICATION_OF_DISKETTE<                         Interim CRC Value = 12821
Line   8  >:UU<                                                Interim CRC Value = 50907
Line   9  >CREATION_DATE<                                      Interim CRC Value =  8815
Line  10  >:<                                                  Interim CRC Value = 16354
Line  11  >CREATION_TIME<                                      Interim CRC Value = 38114
Line  12  >:<                                                  Interim CRC Value = 23188
Line  13  >DISKETTE_RELEASING_OFFICIAL<                        Interim CRC Value = 47545
Line  14  >:<                                                  Interim CRC Value = 41464
Line  15  >ORGANIZATION<                                       Interim CRC Value = 32812
Line  16  >:<                                                  Interim CRC Value = 52737
Line  17  >COMMENTS<                                           Interim CRC Value = 54332
Line  18  >:<                                                  Interim CRC Value =   596
Line  19  >MESSAGES_FOLLOW<                                    Interim CRC Value = 30560
Line  20  >T_MS1WS4.ASC DD173 RR UU    REDY 63619<             Interim CRC Value = 44797

After Final Pass, CRC value = >44797<
```

40.2.4.2 CRC listing of Sample DD 173.

The following CRC listing is included to facilitate checking the CRC output of the developed application software. The message should be entered into the computer as shown, and must be saved to disk as an ASCII text file. The line numbers and arrows mean the same as explained above.

```
CRC LISTING OF SAMPLE DD 173
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
ENTER FILENAME : sample.msg

Line   1  >FORMAT_NAME<                        Interim CRC Value =  9506
Line   2  >:DD173<                             Interim CRC Value =  2733
Line   3  >INTERNAL_MESSAGE_CRC<               Interim CRC Value = 50725
Line   4  >:<                                  Interim CRC Value = 51335
Line   5  >DISTRIBUTION<                       Interim CRC Value =  7101
Line   6  >:00/43/44<                          Interim CRC Value = 40875
Line   7  >DRAFTERS_NAME<                      Interim CRC Value = 62053
Line   8  >:G.J.HOEHL<                         Interim CRC Value =   647
Line   9  >DRAFTERS_OFFICE<                    Interim CRC Value = 24732
Line  10  >:441B, X2195<                       Interim CRC Value = 62788
Line  11  >RELEASERS_NAME<                     Interim CRC Value =  7387
Line  12  >:W.C. BRYSON<                       Interim CRC Value = 17672
Line  13  >SPECIAL_INSTRUCTIONS<               Interim CRC Value =  8167
Line  14  >:<                                  Interim CRC Value = 23007
Line  15  >PAGE_OF<                            Interim CRC Value = 50009
Line  16  >:01    02<                          Interim CRC Value =  6346
Line  17  >DAY_TIME_ZULU<                      Interim CRC Value = 19922
Line  18  >:061622Z<                           Interim CRC Value = 42977
Line  19  >MONTH<                              Interim CRC Value = 23035
Line  20  >:SEP<                               Interim CRC Value = 21039
Line  21  >YEAR<                               Interim CRC Value = 57627
Line  22  >:88<                                Interim CRC Value = 18442
Line  23  >PRECEDENCE_FIRST<                   Interim CRC Value = 45282
Line  24  >:RR<                                Interim CRC Value = 26185
Line  25  >PRECEDENCE_SECOND<                  Interim CRC Value = 48305
Line  26  >:<                                  Interim CRC Value = 26620
Line  27  >CLASS<                              Interim CRC Value = 58928
Line  28  >:UU<                                Interim CRC Value =  3477
Line  29  >SPECAT<                             Interim CRC Value = 56361
Line  30  >:<                                  Interim CRC Value = 52637
Line  31  >LMF<                                Interim CRC Value =  2992
Line  32  >:AC<                                Interim CRC Value = 31575
Line  33  >CIC<                                Interim CRC Value = 38164
Line  34  >:<                                  Interim CRC Value =  7189
Line  35  >OMI<                                Interim CRC Value = 30172
Line  36  >:<                                  Interim CRC Value = 35572
Line  37  >BOOK<                               Interim CRC Value = 29250
Line  38  >:NO<                                Interim CRC Value = 60880
Line  39  >MSG_HANDLING<                       Interim CRC Value =  8795
Line  40  >:ADMIN<                             Interim CRC Value = 15103
Line  41  >FROM<                               Interim CRC Value = 34128
Line  42  >:NAVTASC CHELTENHAM MD<             Interim CRC Value = 19210
Line  43  >TO<                                 Interim CRC Value = 41785
Line  44  >:COMNAVTELCOM WASHINGTON DC<        Interim CRC Value = 31083
```


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```
Line  45  >SPAWARSYSCOM WASHINGTON DC<                    Interim CRC Value = 11818
Line  46  >INFO<                                         Interim CRC Value = 52845
Line  47  >:CNO WASHINGTON DC<                           Interim CRC Value = 60833
Line  48  >NAVCAMS LANT NORFOLK VA<                      Interim CRC Value = 60562
Line  49  >NAVCAMS MED NAPLES IT<                        Interim CRC Value = 16653
Line  50  >NAVCAMS EASTPAC HONOLULU HI<                  Interim CRC Value = 39323
Line  51  >NAVCAMS WESTPAC GUAM<                         Interim CRC Value = 63495
Line  52  >XMT<                                          Interim CRC Value = 32071
Line  53  >:<                                            Interim CRC Value =  8637
Line  54  >ACCT<                                         Interim CRC Value = 16801
Line  55  >:<                                            Interim CRC Value = 43776
Line  56  >FL12A<                                        Interim CRC Value = 58427
Line  57  >:UNCLAS<                                      Interim CRC Value =  4551
Line  58  >FL12B<                                        Interim CRC Value = 11499
Line  59  >:<                                            Interim CRC Value = 23788
Line  60  >FL12B_COMMENT<                                Interim CRC Value = 20748
Line  61  >:<                                            Interim CRC Value =  5841
Line  62  >FL12C<                                        Interim CRC Value = 23394
Line  63  >:<                                            Interim CRC Value = 64090
Line  64  >FL12D<                                        Interim CRC Value =  5811
Line  65  >://N02319//<                                  Interim CRC Value =  3001
Line  66  >FL12E<                                        Interim CRC Value = 64645
Line  67  >:<                                            Interim CRC Value = 45245
Line  68  >TEXT_LINES_FOLLOW<                            Interim CRC Value = 20437
Line  69  >1. THIS IS A SAMPLE MESSAGE PROVIDED FOR THE BENEFIT OF ALL SYSTEM < Interim CRC Value =  8204
Line  70  >DEVELOPERS WHO NEED A WAY TO TEST THEIR DEVELOPED APPLICATION AND <  Interim CRC Value =  3320
Line  71  >THE IMPLEMENTATION OF THE CRC ALGORITHM.  THE RESULTING NUMBER FOR < Interim CRC Value = 18377
Line  72  >EACH LINE OF THIS MESSAGE IS INCLUDED IN THE CRC PRINTOUT PROVIDED < Interim CRC Value = 27620
Line  73  >WITH THIS SAMPLE MESSAGE.  DEVELOPERS MAY USE ONE OF THE PROGRAMS <  Interim CRC Value = 49434
Line  74  >PROVIDED IN THIS APPENDIX, OR MAY DEVELOP THEIR OWN PROGRAM.  <      Interim CRC Value =  1274
Line  75  >HOWEVER, WHATEVER IS USED SHOULD PROVIDE THE SAME NUMERIC RESULTS <  Interim CRC Value = 54366
Line  76  >FOR EACH LINE AS WELL AS FOR THE ENTIRE FILE.  TO RECREATE THIS <    Interim CRC Value = 20594
Line  77  >SAMPLE MESSAGE, IT SHOULD BE TYPED AS SHOWN AS A PURE TEXT FILE.  <  Interim CRC Value = 28288
Line  78  >THERE SHOULD BE NO MARGINS AND ALL LINES SHOULD START IN COLUMN 1 <  Interim CRC Value = 11185
Line  79  >WITH THE TOP OF FILE STARTING ON LINE 1 AFTER THE MESSAGE HAS BEEN < Interim CRC Value = 29030
Line  80  >SAVED TO DISK.  DEVELOPERS MAY ALSO ENTER THE MESSAGE VIA THEIR <    Interim CRC Value = 57349
Line  81  >DEVELOPED APPLICATION.  IN THE CASE OF THE LATTER, THE FILE SAVED TO Interim CRC Value =  7989
Line  82  >DISK SHOULD LOOK LIKE THIS FILE, WITH THE ABOVE COLUMN AND LINE <    Interim CRC Value = 62982
Line  83  >CONSTRAINTS.<                                                        Interim CRC Value = 32433
Line  84  >2. THE LINE-BY-LINE PRINTOUT WAS DONE USING THE PASCAL PROGRAM CODE. Interim CRC Value = 40640
Line  85  >THE DEBUGGING LIST PROVIDES EIGHT (8) LINES OF TRACE VALUES FOR THE <Interim CRC Value = 61267
Line  86  >INTERNAL VARIABLES IN THE CRC CALCULATION ROUTINE FOR ONE (1) BYTE. <Interim CRC Value = 25276
Line  87  >THIS IS PROVIDED TO FACILITATE DEBUGGING VARIOUS IMPLEMENTATIONS OF <Interim CRC Value = 35811
Line  88  >THE CRC ALGORITHM.<                                                  Interim CRC Value = 23031

After Final Pass, CRC value = >23031<
```


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40.2.4.3 DEBUGGING LIST.

The following list provides debugging information on the first byte of the Diskette TOC file, and the Message file. It may be used by system developers as an aid in debugging your program code.

Acc1 =	0	Acc2 =	0			Ch_Ord	= 35	Shf 1	= 0	Shf 2	= 0
Acc1 =	0	Acc2 =	0	Acc3 =	-24575	Ch_Ord	= 17	Shf 1	= 1	Shf 2	= 0
Acc1 =	-24575	Acc2 =	20480			Ch_Ord	= 8	Shf 1	= 1	Shf 2	= 1
Acc1 =	20480	Acc2 =	10240			Ch_Ord	= 4	Shf 1	= 0	Shf 2	= 0
Acc1 =	10240	Acc2 =	5120			Ch_Ord	= 2	Shf 1	= 0	Shf 2	= 0
Acc1 =	5120	Acc2 =	2560			Ch_Ord	= 1	Shf 1	= 0	Shf 2	= 0
Acc1 =	2560	Acc2 =	1280	Acc3 =	-23295	Ch_Ord	= 0	Shf 1	= 1	Shf 2	= 0
Acc1 =	-23295	Acc2 =	21120	Acc3 =	-3455	Ch_Ord	= 0	Shf 1	= 0	Shf 2	= 1
Acc1 =	-3455	Acc2 =	31040			Ch_Ord	= 39	Shf 1	= 1	Shf 2	= 1

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APPENDIX B

TOC HEADER FORMAT

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION.

40.1 Examples. This appendix contains an example of TOC header formats.

```
FORMAT_NAME                                (M)
:TABLE OF CONTENTS/V2
TOC_CRC                                    (M)
:40193
ORIGINATING_SYSTEM                         (M)
:SARAHv7.01
CLASSIFICATION_OF_DISKETTE                 (M)
:UU
CREATION_DATE                              (O)
:890901
CREATION_TIME                              (O)
:07301578
DISKETTE_RELEASING_OFFICIAL                (M)
:GEORGE BOSS, GM13
ORGANIZATION                               (M)
:TOP NOTCH SQUADRON, TNS/ME, 123-4567
COMMENTS                                   (O)
:
MESSAGES_FOLLOW                            (M)
```

(M) - MANDATORY (O) - OPTIONAL

(Indicators (M) and (O) are not part of actual TOC Header.)

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APPENDIX C

DD173 PROLOGUE EXAMPLE WITH ALL OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION.

40.1 Examples. This appendix contains an example of a prologued DD173 message with a card output LMF using all mandatory and optional message information:

```
-------------------------------------------------------------------------
FORMAT_NAME
:DD173
LMF
:AC
INTERNAL_MESSAGE_CRC
: 2478
COORDINATION_MEMO
:COORDINATION REQUIRED
:MFR
:FROM:  CO (Mr Bob Dowright)
:   These  are the comments provided to correct
:   or change this message.  You can enter as
:   many lines of input as needed to convey the
:   information.
:
DISTRIBUTION
:EID/CC - 2
DRAFTERS_NAME
:SSGT I. M. Fine
DRAFTERS_OFFICE
:XPT/45410
RELEASERS_NAME
:BOSS/OIC/CSD/42925
SPECIAL_INSTRUCTIONS
:CONFIRM DELIVERY TO ACTION ADDRESS
:TCC. SEND AS A SINGLE ADDRESS
```


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```
:MESSAGE
PAGE_OF
:01    01
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
PRECEDENCE_SECOND
:RR
CLASS
:UU
SPECAT
:
CIC
:ZZEZ
OMI
:XPP 031425 X
BOOK
:NO
MSG_HANDLING
:ZYB
FROM
:CCSO TINKER AFB OK//SKAA//
TO
:HQ MAC SCOTT AFB IL//DO/LG//
:COLLECTIVE NUMBER 21
INFO
:SSC GUNTER AFB AL//DO//
PLA_RI
:RUWTSUU
XMT
:HQ MAC SCOTT AFB IL
ACCT
:ARX GMRC
FL12A
:UNCLAS
FL12B
:
FL12B_COMMENT
:AMPLIFYING INFORMATION AS ALLOWED BY APPROPRIATE SERVICE PUBLICATIONS
FL12C
:INFORMATION ALLOWED BY PARA 355 ACP 121 US SUPP-1
```


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```
FL12D
:SSIC CODE AND INFORMATION.
FL12E_H
:SPECIAL DELIVERY INSTRUCTIONS
:EXERCISE NAME
:SEE PARA. 323.b ACP 121-1
:REFERENCES
DECLASSIFICATION_INSTRUCTIONS
:DECLASSIFY AFTER 120 DAYS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX D

DD173 PROLOGUE EXAMPLE WITH NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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APPENDIX D

40.1 Examples. This appendix contains an example of a prologued DD173 message with a card output LMF using only mandatory message information:

```
------------------------------------------------------------------------
FORMAT_NAME
:DD173
LMF
:AC
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
FROM
:CCSO TINKER AFB OK//SKAA//
TO
:HQ MAC SCOTT AFB IL//DO/LG//
:COLLECTIVE NUMBER 21   (FOR DD173 FORMAT)
FL12A
:UNCLAS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX E

JANAP 128 NARRATIVE WITH ALL OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION.

40.1 Examples. This appendix contains an example of a prologued JANAP 128 message with a narrative output LMF using normal (plaindress) using all mandatory and optional message information:

```
---------------------------------------------------------------------
FORMAT_NAME
:NR128
LMF
:CA
C_MESSAGE
:N
INTERNAL_MESSAGE_CRC
:90553
COORDINATION_MEMO
:COORDINATION COMPLETE
:MFR
:FROM:  CO (Mr Bob Dowright)
:   These are the comments provided to correct
:   or change this message.  You can enter as
:   many lines of input as needed to convey the
:   information.
:
DRAFTERS_NAME
:SSGT I. M. Fine
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
```


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```
PRECEDENCE_FIRST
:PP
PRECEDENCE_SECOND
:RR
CLASS
:UU
SPECAT
:
CIC
:ZYUW
OSRI_PLA
:RUWTAAA/CCSC TINKER AFB OK
:       /    CONTINUATION LINE STARTS HERE
TO
:RUWTXZZ/HQ MAC SCOTT AFB IL//DO/LG//
:       /    CONTINUATION LINE STARTS HERE
:       /COLLECTIVE NUMBER 22   (FOR J128 AND DD1392)
:RBWTABC/
:RUWTABB/
:RUWTACC/UNIT NAME PLA
INFO
:RUDXXAB/NOC COMM GRP USA
XMT
:UNIT NAME PLA
ACCT
:ARX GMRC
TRC
:BB
FL4_TEXT
:ZFD RUWTABC
:RUWTABC T SOKO SASEBO
FL12A
:UNCLAS
FL12B
:
FL12B_COMMENT
:AMPLIFYING INFORMATION AS ALLOWED BY APPROPRIATE SERVICE PUBLICATIONS
FL12C
:INFORMATION  ALLOWED  BY PARA 355 ACP  121  US SUPP-1
FL12D
:SSIC CODE AND INFORMATION.
FL12E_H
:SPECIAL DELIVERY INSTRUCTIONS
:EXERCISE NAME
:SEE PARA. 323.b ACP 121-1
```


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APPENDIX E

```
:REFERENCES
TIME_OF_FILE
:0761315
ORIGINAL_STATION_SERIAL_NUMBER
:0233
DECLASSIFICATION_INSTRUCTIONS
:DECLASSIFY AFTER 120 DAYS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX F

JANAP 128 NARRATIVE WITH NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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Check the source to verify that this is the current version before use.

APPENDIX F

40.1 Examples. This appendix contains an example of a prologued JANAP 128 message with a narrative output LMF using normal (plaindress) using only mandatory message information:

```
------------------------------------------------------------------------
FORMAT_NAME
:NR128
LMF
:CA
C_MESSAGE
:S
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
CIC
:ZYUW
OSRI_PLA
:RUWTAAA/CCSC TINKER AFB OK
TO
:RUWTXZZ/HQ MAC SCOTT AFB IL//DO/LG//
:       /    CONTINUATION LINE STARTS HERE
FL12A
:UNCLAS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
DECLASSIFICATION INSTRUCTIONS MUST BE INCLUDED IF REQUIRED IN
TEXT HERE.
(((END OF FILE MARK)))
```


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APPENDIX G

DD 1392 WITH RIs AND ALL OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION.

40.1 Examples. This appendix contains an example of a prologued DD 1392 service message with a narrative output LMF using all mandatory and optional message information:

```
---------------------------------------------------------------------
FORMAT_NAME
:P1392
LMF
:CA
C_MESSAGE
:N
INTERNAL_MESSAGE_CRC
: 2478
COORDINATION_MEMO
:COORDINATION REQUIRED
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
DRAFTERS_NAME
:
CIC
:ZYVW
OSRI_PLA
:RUWTAAA/CCSC TINKER AFB OK
```


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APPENDIX G

```
TO
:RUWTXZZ/HQ MAC SCOTT AFB IL//DO/LG//
:       /    CONTINUATION LINE STARTS HERE
:       /COLLECTIVE NUMBER 22   (FOR J128 AND DD1392)

:RBWTABC/
:RUWTABB/
:RUWTACC/
TRC
:BB
FL12A
:UNCLAS
FL12B
:
FL12B_COMMENT
:SVC ZUI SVC 0245 RUEDCSA1234 1921600.
TIME_OF_FILE
:0761315
ORIGINAL_STATION_SERIAL_NUMBER
:0233
DECLASSIFICATION_INSTRUCTIONS
:DECLASSIFY AFTER 120 DAYS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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Check the source to verify that this is the current version before use.

APPENDIX H

DD 1392 WITH RIs AND NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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APPENDIX H

40.1 Examples. This appendix contains an example of a prologued DD 1392 service message with a narrative output LMF using only mandatory message information:

```
------------------------------------------------------------------------
FORMAT_NAME
:P1392
LMF
:CA
C_MESSAGE
:S
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
CIC
:ZYVW
OSRI_PLA
:RUWTAAA/CCSC TINKER AFB OK
TO
:RUWTAAA/CCSC TINKER AFB OK
:       /    CONTINUATION LINE STARTS HERE
:       /COLLECTIVE NUMBER 22
:RUWTABC/
:RUWTABB/
:RUWTACC/
FL12A
:UNCLAS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
DECLASSIFICATION INSTRUCTIONS IF REQUIRED SHOULD BE HERE IN
TEXT.
(((END OF FILE MARK)))
```


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APPENDIX I

JANAP 128 PILOT WITH ALL OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

40.1 Examples. This appendix contains an example of a prologued piloted JANAP 128 message with a card output LMF with all mandatory and optional message information:

```
-------------------------------------------------------------------------
FORMAT_NAME
:PL128
LMF
:CC
C_MESSAGE
:P
INTERNAL_MESSAGE_CRC
: 2278
COORDINATION_MEMO
:COORDINATION REQUIRED
DRAFTERS_NAME
:SSGT I. M. Fine
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
CIC
:ZZEZ
OSRI_PLA
:RUWTAAA/CCSC TINKER AFB OK
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
TO
:RUWTXZZ/HQ MAC SCOTT AFB IL//DO/LG//
:       /    CONTINUATION LINE STARTS HERE
:       /COLLECTIVE NUMBER 22   (FOR J128 AND DD1392)
:RBWTABC/
:RUWTABB/
:RUWTACC/
TRC
:BB
FL4_TEXT
:ZFD RUWTABC
:ZUI SVC 0147 ZDK RUWTAAA 1305 3552316
P_CIC
:ZDKW
P_ADDRESSEE_RI
:RUWTAAA/HQ SSC GUNTER AFB AL//SSMT/
:       /    CONTINUATION LINE STARTS HERE
TIME_OF_FILE
:0761315
ORIGINAL_STATION_SERIAL_NUMBER
:0233
DECLASSIFICATION_INSTRUCTIONS
:DECLASSIFY AFTER 120 DAYS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


Source: https://assist.dla.mil -- Downloaded: 202610-26T07:54Z
Check the source to verify that this is the current version before use.

APPENDIX J

JANAP 128 PILOT WITH NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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Check the source to verify that this is the current version before use.

APPENDIX J

40.1 Examples. This appendix contains an example of a prologued piloted JANAP 128 message with a narrative output LMF with only mandatory message information:

```
------------------------------------------------------------------------
FORMAT_NAME
:PL128
LMF
:CA
C_MESSAGE
:P
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
CIC
:ZZEZ
OSRI_PLA
:RUWTAAA/CCSC TINKER AFB OK
TO
:RUWTXZZ/HQ MAC SCOTT AFB IL//DO/LG//
P_CIC
:ZDKW
P_ADDRESSEE_RI
:RUWTAAA/HQ SSC GUNTER AFB AL//SSMT//
FL12A
:UNCLAS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
DECLASSIFICATION INSTRUCTIONS IF REQUIRED SHOULD BE HERE
IN TEXT.
(((END OF FILE MARK)))
```


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APPENDIX K

JANAP 128 SINGLE CARD

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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Check the source to verify that this is the current version before use.

APPENDIX K

40.1 Examples. This appendix contains an example of a prologued single card JANAP 128 message with all mandatory and optional message information:

```
-----------------------------------------------------------------------
FORMAT_NAME
:DT128
LMF
:SC
C_MESSAGE
:S
INTERNAL_MESSAGE_CRC
: 7478
COORDINATION_MEMO
:MFR
:FROM:  CO (Mr Bob Dowright)
:   These  are the comments provided to correct
:   or change this message.  You can enter as many
:   lines of input as needed to convey the
:   information.
:
DRAFTERS_NAME
:SSGT I. M. Fine
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
CIC
:ZYVW
OSRI_PLA
:RUWTAAA/
TO
:RUWTXZZ/
TEXT_LINES_FOLLOW
THIS IS A 40 CHARACTER MESSAGE AREA....
(((END OF FILE MARK)))
```


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APPENDIX L

JANAP 128 SINGLE CARD AND NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION.

40.1 Examples. This appendix contains an example of a prologued single card JANAP 128 message with only mandatory message information:

```
-------------------------------------------------------------------------
FORMAT_NAME
:DT128
LMF
:SC
C_MESSAGE
:S
PRECEDENCE_FIRST
:PP
CLASS
:UU
SPECAT
:
CIC
:ZYVW
OSRI_PLA
:RUWTAAA/
TO
:RUWTXZZ/
TEXT_LINES_FOLLOW
THIS IS A 40 CHARACTER MESSAGE AREA....
(((END OF FILE MARK)))
```


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APPENDIX M

DSSCS DD173 PROLOGUE FIELD WITH ALL OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page).

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APPENDIX M

40.1 Examples. This appendix contains an example of a prologued DD173 message with all mandatory and optional message information:

```
----------------------------------------------------------------------
FORMAT_NAME
:DD173
LMF
:AA
INTERNAL_MESSAGE_CRC
: 2478
DIA_MSG_LEVEL_CRC
:1EB7
COORDINATION_MEMO
:COORDINATION REQUIRED
:MFR
:FROM:  CO (Mr Bob Dowright)
:   These are the comments provided to correct
:   or change this message.  You can enter as
:   many lines of input as needed to convey the
:   information.
:
DISTRIBUTION
:EID/CC - 2
DRAFTERS_NAME
:SSGT I. M. FINE
DRAFTERS_OFFICE
:XPT/45410
RELEASERS_NAME
:BOSS/OIC/CSD/42925
SPECIAL INSTRUCTIONS
:CONFIRM DELIVERY TO ACTION ADDRESS
:TCC. SEND AS A SINGLE ADDRESS
:MESSAGE
PAGE_OF
:01    01
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
PRECEDENCE_SECOND
:RR
```


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```
CLASS
:EE
SPECAT
:
CIC
:ZYVW
OMI
:XPP 031425 X
BOOK
:YES
MSG_HANDLING
:ZYQ DDI:XXX YYY
FROM
:SITE 1//SKAA//
TO
:SITE 2
:DWXYZ  (NOTE: THIS IS A DUMMY DSSCS ADDRESS GROUP)
INFO
:SITE 7
PLA_RI
:YXXXXX
LESS
:SITE 5
FL12A
:UNCLAS E F T O
CODEWORDS
:XXXXXXX YYYYY ZZZZZZZ AAAAAAAA
CAVEATS
:HANDLE VIA XYXYXYXYX CHANNELS ONLY
DECLASSIFICATION_INSTRUCTIONS
:DECLASSIFY OADR
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX N

DSSCS DD173 PROLOGUE FIELD WITH NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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APPENDIX N

40.1 Examples. This appendix contains an example of a prologued DD173 message with only mandatory message information:

```
------------------------------------------------------------------------
FORMAT_NAME
:DD173
LMF
:AA
INTERNAL_MESSAGE_CRC
: 2478
DIA_MSG_LEVEL_CRC
:1EB7
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
CLASS
:EE
SPECAT
:
OMI
:XPP 031425 X
FROM
:SITE 1//SKAA//
TO
:SITE 2
:DWXYZ  (NOTE: THIS IS A DUMMY DSSCS ADDRESS GROUP)
FL12A
:UNCLAS E F T O
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX O

DOI-103 NARRATIVE WITH ALL OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page.)

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APPENDIX O

40.1 Examples. This appendix contains an example of a prologued DOI-103 message with a narrative output LMF using normal (plaindress) with all mandatory and optional message information:

```
-----------------------------------------------------------------------
FORMAT_NAME
:NR103
LMF
:AA
C_MESSAGE
:N
INTERNAL_MESSAGE_CRC
:90553
DIA_MSG_LEVEL_CRC
:1ED7
COORDINATION_MEMO
:COORDINATION COMPLETE
:MFR
:FROM:  CO (Mr Bob Dowright)
:   These  are the comments provided to correct
:   or change this message.  You can enter as
:   many lines of input as needed to convey the
:   information.
:
DRAFTERS_NAME
:SSGT I. M. Fine
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
PRECEDENCE_SECOND
:RR
CLASS
:EE
SPECAT
:
CIC
:ZYUW
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
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```
OSRI_PLA
:         SITE 1
:             CONTINUATION LINE STARTS HERE
TO
:         SITE 2//INS//
:             CONTINUATION LINE STARTS HERE
:         DSSCS ADDRESS GROUP
:         SITE X1
INFO
:         SITE Q1
LESS
:SITE X1
TCC
:NSH
FL4_TEXT
:ZFD YKKKKK
FL12A
:UNCLAS EFTO
CODEWORDS
:XXXXXX YYYYYYY
CAVEATS
:HANDLE VIA XYXYXYXY CHANNELS
FL12E_H
:SPECIAL DELIVERY INSTRUCTIONS
TIME_OF_FILE
:0761315
ORIGINAL_STATION_SERIAL_NUMBER
:0233
DECLASSIFICATION_INSTRUCTIONS
:DECLASSIFY AFTER 120 DAYS
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX P

DOI-103 NARRATIVE WITH NO OPTIONS

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. (Examples on next page).

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Check the source to verify that this is the current version before use.

APPENDIX P

40.1 Examples. This appendix contains an example of a prologued DOI-103 message with a narrative output LMF using normal (plaindress) with only mandatory message information:

```
------------------------------------------------------------------------
FORMAT_NAME
:NR103
LMF
:AA
C_MESSAGE
:N
INTERNAL_MESSAGE_CRC
:90553
DIA_MSG_LEVEL_CRC
:1ED7
DAY_TIME_ZULU
:010817Z
MONTH
:MAR
YEAR
:89
PRECEDENCE_FIRST
:PP
CLASS
:EE
SPECAT
:
CIC
:ZYUW
OSRI_PLA
:       SITE 1
TO
:       SITE 2//INS//
:           CONTINUATION LINE STARTS HERE
TCC
:NSH
FLI2A
:UNCLAS EFTO
TEXT_LINES_FOLLOW
THIS IS THE TEXT OF A MESSAGE AND NOTICE THAT
IT IS NOT REQUIRED TO BE PRECEDED BY A COLON.
(((END OF FILE MARK)))
```


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APPENDIX Q

DIA MESSAGE LEVEL CRC

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. This section is not applicable to this appendix.

40. DETAILED INFORMATION. The DIA CRC provides a means to ensure a very high level of data integrity during the message handling process. A standard CRC-16 algorithm is used for error detection and CRC validation is performed on all of the following message types:

JANAP 128 DOI 103 DOI 103M ACP 127 Single Card DD Form 173

a. On DIA accredited computer systems which process record communications, the appearance of the DIA CRC is optional as FL15A. The use of FL15A is a line class marking parameter within the computer system. This means that if an input line requires FL15A, messages received over that line must have a FL15A present. If FL15A is present on a message it will be verified regardless of the line markings. On output, messages being transmitted will include FL15A if the output line requires it.

b. The DIA CRC utilizes the CRC-16 algorithm. Following is the pseudo code for the algorithm:

```
procedure CRC_Buffer  (IN Buffer; IN Length; IN OUT CRC_Accumulator) is
  begin
    Poly :=120001(8);
    for i :=1 to length loop
      Letter :=Buffer(i)
      for Bit :=1 to 8 loop
        Shift1 :=Letter rem 2;
        Shift2 :=CRC_Accumulator rem 2;
        CRC_Accumulator :=CRC_Accumulator / 2;
        if Shift1 /=Shift 2
          then
```


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```
            CRC_Accumulator :=CRC_Accumulator XOR Poly;
          end if;
          Letter :=Letter / 2;
        end for;
      end for;
    end CRC_Buffer
```

c. A CRC value is accumulated on FLs 5 (precedence and DTG line), 6 (FROM line), and 12 (Classification and text lines). However, FL12 is the only FL required on all message types.

d. DOI 103 narrative and data pattern, JANAP 128 narrative and data pattern, DOI 103M, and ACP 127 messages are all similar in the following ways: FL5 starts with a precedence and ends with a DTG and may include communication operating signals, FL6 starts with "FM" or "FROM" and is followed by the originator, and FL12 which starts with the classification line and ends with, but does not include, FL15 (pound sign "#", immediately followed by the four digit SSN). However, the similarity ends when it comes to the location of FL15A.

e. For DOI 103 narrative, JANAP 128 narrative, DOI 103M, and ACP 127 messages, FL15A can be found on the line directly after FL15. FL15A is then followed by two CRs, eight LFs, and the EM marked "NNNN", (figures 1, 3, 6, and 7).

f. For DOI 103 data and JANAP 128 data messages, FL15A will be found in columns 73 through 76 of the trailer card (figures 2 and 4).

g. Single card messages are different from other data pattern messages in that the entire card is CRC calculated. Before CRC is calculated, FL15A, columns 5 through 8, is replaced by the correct CIC of "ZYVW" (figure 5).

h. For DD Form 173 messages, the DTG occupies columns 9 through 24 of the first line. A CRC value should be calculated on that field regardless of its contents. FL6 is the third line of the message and may contain "FM" or FROM". FL12 starts with the classification and ends with, but does not include, FL15 and is terminated by two CRs, one LF, and the EM marker "NNNN" (figure 8). CRC calculations are performed on all characters except CRs, LFs, and EM (31(8)) characters.

i. Figures 1 through 8 are examples of messages with FL15a. Data lines used for CRC calculations are underlined and FL15A is in bold type.

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```
RATMZYUW YYYYYY 0003 0011200-MNSH--YYYYYX.
ZNY MMNSH
ZKZK RR SOA DE
R 230600Z APR 89
FM LOCATION A
TO LOCATION B
ZEM
UNCLAS
QQQQ
THIS IS A DOI 103 TEST MESSAGE.
#0003
FFFF
NNNN
```

Figure 1. DOI 103 (Narrative)

```
PCCMZYUW YYYYYY 0004 0071234 0012-MNSH--YYYYYX.
ZNY MMNSH
ZKZK PP SOA DE
P 230600Z APR 89
FM LOCATION 1
TO LOCATION 2
ZEM
UNCLAS
QQQQ
THIS IS A DOI 103 CARD TEST MESSAGE.
PCCMZYUW YYYYYY 0004 071234 0012-MNSH                          FFFFNNNN

Note: The last character of the last card is in column 80.
```

Figure 2. DOI 103 (Data Pattern)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
ZATEZYUW RUCSSVC0002 0051300-EEEE--RUCSMDC.
ZNY EEEEE
Z 230600Z APR 89
FM SECDEF WASHINGTON DC
TO HQ USAF//CC//
BT
UNCLAS E F T O
SUBJ:  JANAP 128
THIS IS A JANAP 128 TEST MESSAGE.
BT
#0002
FFFF




NNNN
```

Figure 3. **JANAP 128 (Narrative)**

```
PACEZYUW RUCSSVC0003 0081217 0100-EEEE--RUCSMDC.
ZNY EEEEE
P 230600Z APR 87
FM SECDEF WASHINGTON DC
TO HQ USAF//CC//
BT
UNCLAS EFTO
SUBJ: JANAP 128 CARD
THIS IS A JANAP 128 CARD TEST MESSAGE.
PACEZYUW RUCSSVC0003 0081217 0010-EEEE                          FFFFNNNN

Note:  The last character on the last card is in column 80.
```

Figure 4. **JANAP 128 (Data Pattern)**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
Check the source to verify that this is the current version before use.

```
PSCUFFFF RUCSSVC005 0312100-RUCSMDC. SINGLE CARD EXAMPLE                NNNN

Note:  the  last character on the card is in column 80.  Also, the  FL15A  is
replaced by the correct CIC, ZYVW, when the CRC is calculated.
```

Figure 5. JANAP 128 (Single Card Data Pattern)

```
RR YYYYYY
DE YYYYYX #0006 1131200
ZNY MMNSH
ZKZK RR SOA DE
R 230600Z APR 89 ZYH
FM SITE 1
TO SITE 2
ZEM
UNCLAS
QQQQ
THIS IS A DOI 103M TEST MESSAGE
#0006
FFFF




NNNN
```

Figure 6. DOI 103M (Narrative)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:54Z
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```
OO RUCSOFA
DE RUCSSVC #007 1141200
ZNR UUUUU
O 230600Z APR 89
FM LOCATION 1
TO LOCATION 2
BT
UNCLAS
SUBJ: ACP 127
THIS IS A ACP 127 TEST MESSAGE.
BT
#0007
FFFF




NNNN
```

Figure 7. ACP 127 (Narrative)

```
01  01  230600Z  APR  89  PP  RR  UUUU      AT      MSGIDENT

LOCATION 1
LOCATION 2
INFO LOCATION 3
UNCLAS
QQQQ
SUBJ: THIS IS A TEST
THIS IS A DD FORM 173 TEST MESSAGE
#0000
FFFF
NNNN
```

Figure 8. DD Form 173

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APPENDIX R

RELEASING DOCUMENT

10. GENERAL

10.1 Scope. This appendix is a required part of the standard. The information contained herein is intended for guidance only. Although considerable effort has been made to assure correctness of the examples in this appendix, it is incumbent upon the reader (user) to validate them prior to use in defense messaging applications.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS. Refer to paragraphs 4.1 and 5.1 for field descriptions.

40. DETAILED INFORMATION. The releasing document will at least contain the following infomation:

UNCLASSIFIED PAGE xx of xx

DISKETTE RELEASING DOCUMENT

DISKETTE VOLUME LABEL: vvvvvvvvvvv

TABLE OF CONTENTS CRC: nnnnn

CLASSIFICATION OF DISKETTE: full classification

COMMUNITY OF DISKETTE: aa

ORGANIZATION/OFFICE/PHONE:

RELEASING OFFICIAL: _____
(name/rank)

Reserved for service/agency use

.
.
.

FILENAME----	TYPE-	PR	CL	SP	ACTN	CRC--	DTG---------	SSN-	TOF/TOR	CIC-
--------.---	-----	--	--	--	----	-----	----------	----	-------	----

.
.
.

UNCLASSIFIED

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